IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Date of publication xxxx 00, 0000, date of current version xxxx 00, 0000.

Digital Object Identifier 10.1109/ACCESS.2024.0429000

Coupling Authorization and Payment Layers in a PageRank Walk for Smart-Contract Reputation: A Pre-Specified, Commit-Gated Out-of-Window Test on an Ethereum Rollup

TAEHONG KWON[1], ERNEST MNKANDLA[1], DONATIEN KOULLA MOULLA[2], AND DAVID SENA ATTIPOE[2]

[1]School of Computing, College of Science, Engineering and Technology, University of South Africa, Johannesburg, South Africa (e-mail: thkwon@enu-tech.co.kr)

[2]Centre for Augmented Intelligence and Data Science (CAIDS), School of Computing, University of South Africa, Johannesburg, South Africa

Corresponding author: Taehong Kwon (e-mail: thkwon@enu-tech.co.kr).

This research received no external funding.

ABSTRACT Blockchain reputation scores usually rank addresses by token payments, yet allowances, through which an owner authorizes a spender to move its tokens, link the same addresses. We treat payments and authorizations as two layers of one network and test on Arbitrum One whether a random walk mixing them at every node ranks smart contracts better than either layer alone. Two decision rules were committed to the authors' public repository before this study extracted their evaluation logs, the second eight seconds before its July to September 2026 window was queried by code that checks the committed files. The main analysis, defined after the plan's results were known and hence a re-analysis, scores 6,483 contracts approved by three or more owners from US-dollar amounts on five tokens in the twelve months before each freeze. Neither rule was met. The coupled walk ranked contracts that later gained approvals better than the payment walk (rank-correlation difference 0.170, 95% interval 0.149 to 0.189), but ranked later payers worse by more than the 0.01 margin (-0.025, interval -0.038 to -0.015), and on the liquidation-free trading of contract traders it was level with the payment walk but too imprecise for the margin. Contrasts computed after the labels placed it below the raw count of approving owners on later approvals in both windows (second window -0.065, interval -0.077 to -0.054). The pre-specified raw-unit analysis reached the same verdicts. In a post hoc model, 26 colluding addresses would lift a target's authorization score above nine tenths of the cohort.

INDEX TERMS Blockchain, decentralized finance, multilayer networks, negative results, PageRank, preregistration, reputation systems, smart contracts, Sybil attacks, token allowances.

I. INTRODUCTION

LENDING and margin protocols in decentralized finance (DeFi) deal with counterparties and integrations that are known only by an address [1]. They rely on over-collateralization and automatic liquidation in place of credit histories [2], [3], and a line of research tries to recover some of the missing information from the public ledger. Most of it ranks addresses on a graph of token transfers. Do and Do read PageRank and HodgeRank on Ethereum transactions as a measure of social credit [4]; Adaptive Weighted PageRank (AWP) of Do, Do and Nguyen weights every transfer by its value and age and restarts the walk at recently active senders [5]; RiskProp propagates account risk through the transaction network [6]. Surveys of blockchain reputation systems find the same reliance on payment volume or on explicit ratings [7]. Anyone can recompute such scores from public events, which answers part of the worry that decentralized credit scoring turns into a black box [8].

A transfer records a payment. The ledger also records a second relation among the same addresses. Under the ERC-20 (Ethereum Request for Comments 20) token interface an owner calls `approve` to let a spender draw up to a stated amount of its tokens, and the `permit` function of Ethereum Improvement Proposal (EIP) 2612 sets the same allowance from an off-chain signature [9], [10]. An allowance moves no funds. It grants a capability that the spender may exercise

later and that the owner may raise, lower or revoke, and many allowances are unlimited [11]. Allowances point mostly from user accounts to the routers, vaults and trading contracts those accounts use, so they are a natural signal for ranking smart contracts. Payments and authorizations thus form two layers over one set of addresses, the setting of multilayer ranking [12], [13].

We ask whether one random walk over both layers ranks contracts better than a walk over either layer alone, and in particular whether adding authorization edges improves on a walk over payments. We define a coupled PageRank (C-PR) that blends the two transition rows of each node with a fixed weight λ on the authorization layer, and an ablation, endorsement-seeded PageRank (S-PR), that walks the payment layer but draws its restarts from the authorization walk, as TrustRank draws its restarts from a vetted seed set [14]. The end points of the C-PR family, $\lambda = 1$ and $\lambda = 0$, walk one layer alone and are the comparators of both decision rules. The scored objects are smart contracts throughout; externally owned accounts enter the graphs as owners, senders and recipients and are never scored.

The evaluation looks forward in time. Scores are frozen at a date and judged by their Kendall rank correlation [15] with outcomes that arrive afterwards: the new owners that approve a contract, the new addresses that pay it, and, for contracts that trade on the GMX V2 perpetual-futures exchange [16], the share of their position closes that escaped forced liquidation. The first two outcomes are arrivals of the same kinds of edge that build the layers; the third is built from neither layer and is the outcome closest to counterparty risk in these data [17].

Two decision rules were fixed before this study extracted the data that judge them. The analysis plan, which fixes Rule A and every score, was committed to the authors' public repository ten seconds before the BigQuery job that copied the observation window was created. The registration of the second window, which fixes Rule B with a non-inferiority margin, was committed eight seconds before the job that copied the logs of July to September 2026, and the script that runs that job refuses to start unless the registration files are committed without local edits; it then records their SHA-256 hashes in a manifest. We call this arrangement commit-gated. The precedence of both commits over the extractions rests on commit and job times, not on a third-party timestamp (Section IV-C). Both rules failed. After these results were known, we found three defects in how the plan weighted amounts and defined the analysis that this paper reports first: amounts in US dollars on five tokens with reliable prices, a cap on unlimited allowances, and scores computed from the twelve months before each freeze. Because under the plan the amounts either had no effect or let unlimited approvals dominate, we lead with this revised analysis; because its definitions came after the results, its verdicts are a re-analysis under the same rules, and the pre-specified analysis, which reached the same verdicts, is reported beside it in every table and figure of results.

The contributions are the following.

1) A two-layer model of contract reputation with a coupled walk, a seeded ablation and two analytic remarks: for λ strictly between 0 and 1 the weight acts only through nodes with out-edges in both layers, and on a graph whose spenders are approved by owners without approvals of their own the authorization walk reduces to an owner-normalized in-strength.
2) A pre-specified, commit-gated protocol for ranking experiments on ledger data, with the record that dates each commitment (Section IV) and the limits of that record: commit times are set by the committer's clock, and no third-party timestamp anchors them.
3) A negative result on Arbitrum One. C-PR met neither rule, in the revised analysis or in the pre-specified one. Its gain over the payment walk on later approvals held in a window that this study extracted after the rule was fixed, but on later payers it fell behind by more than the registered margin, against AWP by more still, and on the trader outcome it did not stay within the margin. In paired contrasts computed after the labels were known, C-PR ranked later approvals below the raw count of approving owners and later payers below the raw count of distinct senders, in both windows of the revised analysis.
4) A post hoc calibration of known link-farm results to the authorization layer under uniform restarts. On the revised EndorseRank graph a closed farm collects 3.23 times its population share at damping 0.85, and a star of 26, about 2,292 or about 47,073 farm addresses lifts its target past the 90th percentile, the 99th percentile or the top of the cohort's EndorseRank scores; on the registered graph the figures are 2.96, 13, 1,253 and about 1,080,271.

A companion manuscript [18] compares the two edge types as separate scores, sets each against its raw degree and studies the trader outcomes, built from neither edge type, under three rules (R1–R3) registered in the same commit as Rule B. Both papers describe the same data and report the spender holdout coefficients of the raw degrees, EndorseRank and AWP with their degree contrasts, the closed form of the authorization walk and the solver runtimes. The allowance-against-transfer comparison on trader outcomes (rules R1–R3 of that study) is reported only there; on the same trader cohort this paper reports the contrasts of C-PR that Rule B and its registered sensitivity analyses name and one contrast of C-PR with the in-approve degree computed afterwards. Both papers report the revised analysis first, with the registered values beside them, and they report the same values for both analyses. This paper concerns the coupling operator, its two pre-specified rules and their record, and the manipulability of the authorization layer. Section II reviews related work, Section III defines the model, Section IV states the rules and their record, Section V describes data and measurement, Section VI reports the results, Section VII calibrates farm attacks, Section VIII discusses threats to validity, Section IX draws implications

for lending and margin protocols, and Section X concludes. Appendix A collects exploratory checks.

II. RELATED WORK

A. REPUTATION FROM LEDGER GRAPHS

PageRank ranks the nodes of a directed graph by the stationary distribution of a damped random walk [19], [20]. Hyperlink-Induced Topic Search (HITS) separates hubs from authorities [21], and EigenTrust normalizes local trust values so that peers in a file-sharing network cannot inflate their own standing [22]. On public ledgers these ideas have been applied almost entirely to transfers. Besides the methods named above [4], [5], network embeddings of transaction graphs detect phishing accounts better than per-account features [23], and a survey of knowledge discovery in cryptocurrency transactions lists degree- and value-based centralities as the usual address descriptors [24]. Approvals appear in the literature in two roles: as a clustering heuristic that links addresses of one entity [25], and as a security risk when unlimited [11]. Deployed tooling such as Human Passport scores the holder of an address from weighted credentials and does not rank relations between addresses [26]. The companion study [18] uses the latest allowance as the edge of a ranking; the present paper combines that edge with the payment edge in one walk.

B. MULTILAYER AND PERSONALIZED RANKING

Multiplex PageRank lets the centrality a node earns in one layer bias the walk in another [12], and the interconnected-multilayer framework of De Domenico *et al.* lets a walker move inside layers and switch between them, ranking nodes by their versatility across layers [13]. Topic-sensitive and personalized PageRank change the teleportation vector rather than the transition matrix [27], [28], and TrustRank confines teleportation to a hand-checked seed set to demote web spam [14]. C-PR is a special case of the first family: a per-node convex combination of two row-stochastic matrices with one global weight and no copies of a node across layers. S-PR belongs to the second family. Proposition 1 below is a first-order perturbation bound of the standard kind for PageRank [20], specialized to this convex combination. We have not found in the literature a test of either construction on two on-chain relations among the same addresses, judged by a rule that the study fixed before it extracted its evaluation data.

C. DEGREE BASELINES

Under preferential attachment new links go disproportionately to nodes that already have many [29], and measurements of growing networks show the same tendency [30]. Degree-based scores are therefore standard baselines in link prediction [31], including on Ethereum transaction networks [32], and on graphs with weak degree correlations PageRank is on average proportional to in-degree [33]. A claim that a walk adds information should thus be tested against the raw degree it smooths, and we set every walk against it.

D. MANIPULATION AND SYBIL RESISTANCE

Where identities are free, one actor can appear as many [34]. Gyöngyi and Garcia-Molina derive the link farms and alliances that raise a target's PageRank the most [35]. Cheng and Friedman show that no symmetric reputation function, PageRank included, is Sybil-proof, whereas some asymmetric functions computed from flows or paths out of a trusted source are [36]. Section VII applies these results to the authorization layer; it calibrates them rather than deriving new ones. On ledgers, wash trading inflates payment activity [37], [38], proof-of-personhood schemes trade Sybil resistance against privacy [39], and a recent review of decentralized systems names credential farming among the remaining weaknesses of such gates [40].

E. PRE-SPECIFIED ANALYSES

Preregistration separates the analyses that test a hypothesis from those that generate one, and it can be applied to data that already exist, provided the analysts have not yet examined the outcomes [41]. With observational ledger data the events of a test window exist on chain from the moment they occur, so the strongest practical guarantee a study can give is that it had not extracted them when its rule was fixed. Construct-validity theory adds a caution that matters below: a score should agree with checks of what it was designed to track, so such agreement is not independent evidence [42].

III. MODEL

A. TWO LAYERS ON ONE SET OF ADDRESSES

Let $V = V_A \cup V_T$ be the set of addresses in either layer (Fig. 1). We use *authorization* for the layer, its edges and its walk, *approval* for the event that creates an edge, and *allowance* for the approved amount; likewise *payment* for the second layer and *transfer* for its events. An authorization edge $u \to v$ exists when the latest `Approval` event from owner u to spender v up to the anchor date carries a positive value on at least one token; a latest value of zero removes the pair. This is what the pipeline measures, and it differs from the allowance in force in three ways: allowances set before the start of the score window and not re-emitted inside it are invisible; a finite allowance that `transferFrom` has since drawn down keeps its approved value, because a token need not emit an `Approval` event when an allowance is spent; and an owner who approves an allowance manager such as Uniswap's Permit2 appears with an edge to that manager only. A payment edge $u \to v$ exists when u sent v at least one transfer with a positive amount. Native ether moves without `Transfer` logs and enters the payment layer only as wrapped ether.

The operator studied here weights both layers by amounts:

$$\tilde{w}_A(u,v) = \sum_{\text{token}} g\big(a^\star(u,v,\text{token})\big), \tag{1}$$

$$\tilde{w}_T(u,v) = \sum_{e:\, u\to v} x_e\, \sigma(\Delta t_e), \tag{2}$$

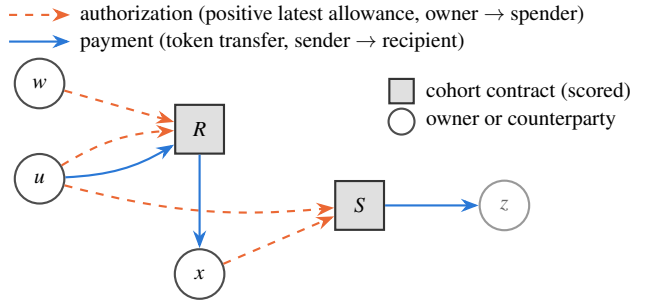


FIGURE 1. The two layers on one set of addresses (schematic). Squares are cohort contracts, the scored objects; circles are owners and counterparties, such as user accounts, and *z* is an address that appears only because a cohort contract paid it. Only *u* has out-edges in both layers, so only its row depends on λ for $0 < \lambda < 1$. At $\lambda = 1$ the payment edges and *z* are dropped, so the contracts *R* and *S* become dangling; at $\lambda = 0$ the authorization edges are dropped, so *w* leaves the graph and *x* becomes dangling. S-PR walks the solid edges only and restarts each jump in proportion to the authorization walk's score.

where $a^\star$ is the latest allowance, x_e the amount of transfer e, and $\sigma(\Delta t) = 1/(1 + \exp(k(\Delta t - t_0)))$ a logistic discount on the age Δt in days, the form used by AWP [5]; that paper fixes no parameter values, and $k = 0.01$ and $t_0 = 180$ are our choices. Ages are measured back from the anchor. The authorization layer is not discounted. The revised analysis of Section IV-D, the main analysis, values amounts in US dollars over five tokens and caps each allowance, $g(a) = \min(a, U)$ with $U = 19{,}000$ dollars; the plan fixed raw token base units, every ERC-20 token and $g(a) = a$. Write $o_A(u)$ and $o_T(u)$ for the out-strengths and P_A, P_T for the row-normalized matrices, whose rows are zero where a node has no out-edge in that layer.

B. PAGERANK WITH A RESTART DISTRIBUTION

Every score below is the fixed point of

$$\mathbf{r}^{(t+1)} = (1-d)\,\boldsymbol{\pi} + d\,P^\top \mathbf{r}^{(t)} + d\,\left(\mathbf{1}_{\text{dang}}^\top \mathbf{r}^{(t)}\right)\boldsymbol{\pi}, \qquad (3)$$

where P is the transition matrix, $\boldsymbol{\pi}$ the restart distribution, and $\mathbf{1}_{\text{dang}}$ the indicator of nodes with an all-zero row, whose mass is sent back through $\boldsymbol{\pi}$. We use $d = 0.85$, start from the uniform vector, and stop when the ℓ_1 change falls below 10^{-8} or after 300 iterations.

C. COUPLED PAGERANK

C-PR mixes the two rows of each node, renormalizing over the layers in which the node has out-edges:

$$P^{(\lambda)}_{u\cdot} = \alpha_A(u)\,P_{A,u\cdot} + \alpha_T(u)\,P_{T,u\cdot}, \qquad (4)$$

$$\alpha_A(u) = \frac{\lambda\,\mathbf{1}[o_A(u) > 0]}{D_\lambda(u)}, \quad \alpha_T(u) = \frac{(1-\lambda)\,\mathbf{1}[o_T(u) > 0]}{D_\lambda(u)}, \qquad (5)$$

with $D_\lambda(u) = \lambda\,\mathbf{1}[o_A(u) > 0] + (1-\lambda)\,\mathbf{1}[o_T(u) > 0]$. A node with out-edges in one layer follows that layer with weight one, and a node with none in either is dangling. Restarts are uniform. The primary setting $\lambda = 0.5$ and the sensitivity values 0.25 and 0.75 were fixed in the plan (Section IV).

At $\lambda = 1$ the payment layer is dropped together with the nodes only it contains, and the operator becomes PageRank on the authorization layer alone; at $\lambda = 0$ it becomes PageRank on the payment layer alone. We call these the authorization walk and the payment walk; they are the comparators of both rules. The family is discontinuous at both ends. For every λ strictly between 0 and 1, a node whose out-edges are all payments, such as a contract that pays out but approves no one, follows the payment layer, whereas at $\lambda = 1$ the same node is dangling or absent. Symmetrically, a node whose out-edges are all authorizations follows the authorization layer at every interior λ but is dangling or absent at $\lambda = 0$. Unit tests on small fixtures check that each end point reproduces its single-layer walk.

D. SEEDED ABLATION

S-PR keeps the payment walk and replaces its uniform restart vector by the authorization walk's score $\mathbf{r}_A$, restricted to V_T and renormalized:

$$\pi_u = \frac{r_{A,u}}{\sum_{v \in V_T} r_{A,v}}, \qquad u \in V_T, \qquad (6)$$

with $r_{A,u} = 0$ for $u \notin V_A$ and a uniform fallback if the denominator vanishes. Authorization then decides where the walker lands after a jump, and payments decide where it goes between jumps. Comparing S-PR with C-PR separates the authorization edge as a route from the authorization edge as a seed.

E. REFERENCE SCORES

Two published or companion scores and two counts serve as references. AWP [5] walks the payment layer with weights $\sigma(\Delta t_e)V(x_e)$ summed over transfers, where $V(z) = 2/(1 + e^{-bz}) - 1$ is a bounded value transform, and restarts at each address in proportion to its recent sending activity. EndorseRank [18] walks the authorization layer with the same bounded weights, applied to the latest allowance and its age, and with uniform restarts. In the revised analysis $b = \ln 3/m$, with m the median amount in the twelve months up to 31 March 2026 rounded to two significant digits, so that an amount of m weighs one half ($m_T = 170$ dollars for transfers, from 169.3, and $m_A = 100$ dollars for finite allowances, from 99.97); in the plan $b = 1$ on raw base units, which makes $V(z)$ within 10^{-4} of one for any amount of ten base units or more. Both reference walks bound the weight of each event, whereas C-PR lets amounts matter without a bound up to the cap. The counts are the in-approve degree, the number of distinct owners holding a positive latest allowance to the contract, and the transfer in-degree, the number of distinct addresses other than itself that sent it a transfer, both taken at the freeze.

F. WHERE THE WEIGHT ACTS

Let $B = \{u : o_A(u) > 0 \text{ and } o_T(u) > 0\}$ be the set of nodes with out-edges in both layers.

Proposition 1: For $\lambda \in (0, 1)$ and uniform $\boldsymbol{\pi}$, the rows of $P^{(\lambda)}$ outside B do not depend on λ, and the derivative $\mathbf{r}'(\lambda)$ of the stationary vector of (3) with respect to λ satisfies

$$\mathbf{r}'(\lambda) = d\,\left(I - d\,G_\lambda^\top\right)^{-1} \sum_{u \in B} r_u \left(\mathbf{p}_{A,u} - \mathbf{p}_{T,u}\right), \qquad (7)$$

where $\mathbf{p}_{A,u}$ and $\mathbf{p}_{T,u}$ are the transposed rows of P_A and P_T and G_λ is $P^{(\lambda)}$ with each dangling row replaced by $\boldsymbol{\pi}^\top$. Hence $\|\mathbf{r}'(\lambda)\|_1 \le 2d\, r(B)/(1-d)$ with $r(B) = \sum_{u \in B} r_u$.

Proof: Outside B one indicator in (5) is zero, so the row is a single layer's row or zero whatever λ is, and the dangling set does not change on $(0, 1)$. For $u \in B$, $D_\lambda(u) = 1$ and the row is $\lambda P_{A,u\cdot} + (1-\lambda) P_{T,u\cdot}$, whose derivative is $P_{A,u\cdot} - P_{T,u\cdot}$. Because $\mathbf{r}$ sums to one, (3) can be written $\mathbf{r} = (1-d)\boldsymbol{\pi} + d\,G_\lambda^\top \mathbf{r}$; differentiating gives (7). The matrix $G_\lambda^\top$ is column-stochastic, so $\|(I - dG_\lambda^\top)^{-1}\|_1 \le 1/(1-d)$, and each difference of two probability rows has ℓ_1 norm at most 2. ∎

The proposition says where the weight can act, not that its action is small: when $r(B)$ is large the bound exceeds the trivial bound of 2 on the ℓ_1 distance between two distributions. In our graphs almost every node of the authorization layer is an owner with out-edges, so every owner that also appears as a sender of an extracted transfer belongs to B. We did not compute $r(B)$; Appendix A reports how far the rankings move between interior values of λ.

G. THE AUTHORIZATION WALK ON DEPTH-ONE SPENDERS

Our graphs keep only events with a cohort contract at one end, so most nodes of the authorization layer either only grant approvals (pure owners) or only receive them (pure spenders). The companion study derives a closed form for this structure [18]; we restate it with its proof.

Proposition 2: Under (3) with uniform $\boldsymbol{\pi}$, every node without in-edges has the same score c, and a node v whose in-neighbors all lack in-edges scores $r_v = c\,(1 + d\,\delta(v))$, where $\delta(v) = \sum_u w(u,v)/o(u)$ is its owner-normalized in-strength.

Proof: With uniform restarts and dangling mass r_D, (3) reads $r_v = (1-d+d\,r_D)/|V| + d\sum_u r_u\, w(u,v)/o(u)$. A node without in-edges receives only the first term, which we call c. If every in-neighbor u of v lacks in-edges, then $r_u = c$, and substituting gives $r_v = c + d\,c\,\delta(v)$. ∎

The statement holds for any edge weights, so it covers EndorseRank and C-PR at $\lambda = 1$, each with its own δ. On the EndorseRank graph at the freeze of 30 June 2026, 5,232 of the 6,224 cohort spenders of the revised analysis have depth one, and the computed score lies within 1% of the closed form for every one of them; over all spenders the share is 87.5%, and 21.9% of the deeper ones [18]. In the registered analysis the figures are 20,942 of 26,512, every one of them, 86.4% and 35.1%. For most spenders the authorization walk is therefore a reweighted count of approving owners, and C-PR inherits that limit wherever it follows the authorization layer.

TABLE 1. Solver Time on the Full Graphs at the Freeze of 30 June 2026 (Mean and SD of Five Timed Runs After One Warm-Up)

Score	$\lvert V\rvert$	$\lvert E\rvert$	Iter.	Time (s)
Revised analysis				
EndorseRank	2,513,982	3,808,949	57	2.66 (0.11)
AWP	3,601,727	8,426,124	94	8.53 (0.62)
C-PR ($\lambda = 0.5$)	4,111,490	10,562,390	82	11.56 (0.21)
S-PR	3,603,300	8,432,635	151	10.88 (0.33)
Registered analysis				
EndorseRank	13,055,150	22,321,470	84	29.60 (0.61)
AWP	23,029,298	59,737,617	93	69.82 (2.30)
C-PR ($\lambda = 0.5$)	25,304,106	71,375,442	92	105.51 (2.48)
S-PR	23,029,298	59,737,617	185	100.86 (4.82)

One session of one warm-up and five timed runs per graph and setting, the measurement cited throughout; the time covers matrix assembly and power iteration ($d = 0.85$, tolerance 10^{-8}, at most 300 iterations) on edge lists built in advance. S-PR includes the solve of the authorization layer that supplies its restart vector. $|V|$ and $|E|$ are those of each solver graph, which drops pairs whose weight is zero in floating point; in the revised analysis this removes dust pairs from the graphs of EndorseRank and AWP (Section V-B). Same machine for both analyses (Section III-H).

H. COST

One iteration of (3) costs $O(|E|)$ [43], and C-PR walks the union of both edge sets. Table 1 reports the solver time of each score on the full graphs at the freeze of 30 June 2026, with $|V|$ and $|E|$ of each solver graph. In the revised analysis C-PR takes 11.56 s on 4,111,490 nodes and 10,562,390 edges, 1.36 times AWP's time on 1.25 times its edges; in the registered analysis 105.51 s on 25,304,106 nodes and 71,375,442 edges, 1.51 times AWP's time on 1.19 times its edges. EndorseRank is the fastest score because its graph is the sparsest: the authorization graph holds 45.2% of the payment graph's edges in the revised analysis and 37.4% in the registered one. The time ratio of AWP to EndorseRank, 3.21 and 2.36, combines that edge ratio (2.21 and 2.68) with the iteration counts of Table 1: AWP needs 94 iterations to EndorseRank's 57 in the revised analysis and 93 to 84 in the registered one. The speed reflects the smaller edge count and the iterations that graph needs, not the algorithm, and any walk that uses the payment layer pays for its edges. Times are means of five runs after one untimed warm-up on one machine (Intel Core Ultra 9 185H, 64 GB of memory, Windows 11, Python 3.11.9, NumPy 2.3.2, SciPy 1.16.1).

IV. PRE-SPECIFIED RULES AND THEIR RECORD

Table 2 lists the commitments in the order in which they were made, the record of each one, and what was known at each point. Three features of this history matter for interpretation. First, Rule A and the coupled operator were fixed before any log of this study was extracted, so no score or label of this study had been computed. Second, the comparison that motivated Rule B was noticed in the first window after its labels were known, so the evidence from that window is exploratory. Third, the revised analysis, with US-dollar

amounts, five tokens and a one-year score window before each freeze, was defined after every result of the pre-specified analysis was known; it applies the same rules and thresholds, but its verdicts are not registered tests. Section VI reports it first, with the registered analysis beside it.

A. RULE A (ANALYSIS PLAN, 9 OCTOBER 2026, 13:41:27 UTC)

Rule A asked whether C-PR beats both layers walked alone. Its primary part used the first holdout window, W0 (scores frozen on 31 March 2026, labels from April to June 2026): at $\lambda = 0.5$, C-PR should do no worse than the authorization walk on new approval pairs, meaning that the 95% interval of $\Delta\tau = \tau_{\text{C-PR}} - \tau_{\lambda=1}$ contains zero or lies above it (A1), and should beat the payment walk on new transfer senders, meaning that the interval of $\tau_{\text{C-PR}} - \tau_{\lambda=0}$ lies above zero (A2). Its secondary part used the matched cohort in the window of the scores themselves: C-PR's point estimate had to lie inside the 95% interval of the better single layer on a family of transfer checks and on a family of authorization checks, and C-PR had to exceed both layers on a family of three Sybil-adjusted stability checks (the plan's name; the share of distinct inbound counterparties, tenure and active months). If the primary part failed, the question whether C-PR beats a walk over either layer alone was to be answered in the negative, and no λ was to be tuned after labels were seen.

Two criteria of the rule are weak by design. ''No worse'' as defined rewards imprecision, since a wider interval contains zero more easily, and the secondary part compares a point estimate with another score's marginal interval rather than using a paired contrast. Anyone reusing the protocol should replace the first by a non-inferiority margin and the second by paired contrasts; we report the paired contrasts beside the rule.

B. RULE B (REGISTRATION, 9 OCTOBER 2026, 16:49:32 UTC)

Rule B asked whether C-PR improves on the payment walk, $\Delta\tau = \tau_{\text{C-PR}} - \tau_{\lambda=0}$, in a second window, W1, frozen on 30 June 2026 with labels counted from 1 July to 30 September 2026:

- F1 (spenders; new approval pairs), superiority: the whole 95% interval of $\Delta\tau$ must be positive;
- F2 (spenders; new transfer senders), non-inferiority: the lower end of the interval must lie above $-\delta$;
- F3 (traders; liquidation-free close rate), non-inferiority with the same margin.

All three had to hold. Because the decision is an intersection of three conditions, testing each at the 5% level keeps the probability of a false claim of improvement at or below 5% without a multiplicity correction. The margin $\delta = 0.01$ is the half-width of the W0 interval of the contrast that F2 repeats (-0.030, $[-0.034, -0.026]$, half-width 0.0040), rounded up to two decimals. It measures the precision of a spender contrast, not a difference judged to matter in practice, and the registration applied it to the trader contrast F3 as well, whose interval rests on far fewer contracts (Section VI-I). The registration also recorded a weakness of F3 in advance: it could pass with both coefficients near zero if neither walk predicted liquidation, so both coefficients are reported beside the verdict.

Three further results were to be reported in a fixed order outside the decision: F4, superiority of C-PR over the payment walk on the liquidation-free close rate; F5, superiority over the authorization walk on that label; and F6, the primary part of Rule A rerun on W1. ''Fixed order'' meant only the order of reporting; there was no gatekeeping, so F4 to F6 are nominal results with unadjusted 95% intervals. Two sensitivity analyses were registered with them: S1 repeats F1 to F4 with AWP as the comparator, and S2 repeats F1 to F6 with cohort contracts that a graph leaves out kept as isolated nodes rather than scored zero. Neither enters the decision. Scores, solver, queries, cohort filter, exchange-event decoder and bootstrap (400 paired resamples, seed 42) were those of the plan.

C. GATE ON EXTRACTION

The configuration carries a status field for W1, which the registration commit set to ''registered''. Before it queries anything, the extraction script checks that the registration and the configuration are committed without local changes and that the status is ''registered''; it then records the commit and the SHA-256 hashes of both files in a manifest. The gate compares no hash; the hashes let a reader check afterwards that the committed files are the ones the script saw. The script has no override, and the manifest's field `override_used` is a constant that records nothing. Table 3 gives the full identifiers of the commits and the hashes. The manifest hashed the files on the authors' Windows checkout, with CRLF line ends; the same text with LF line ends, as git stores it, gives the second pair of hashes, and both were checked against the committed files.

What the record shows depends on who keeps it. The commit times are set by the clock of the machine that made the commits and can be chosen by their author. The BigQuery job times (creation of scan A at 13:41:37 and of scan B at 16:49:40, Table 2) are kept by the cloud provider in the project's job history, which only members of the project can read; we copied them into the data description committed in `f33a87a` (23:24:39 UTC on 9 October). Outsiders can therefore check the content of the registration, the hashes and the order of the commits, but the claim that the registration preceded the W1 extraction rests on the commit and job times, not on a third-party timestamp. A trusted timestamp, such as anchoring the commit hash on chain or in an independent registry before extraction, would remove this gap.

D. THE REVISED ANALYSIS

After the registered results of all windows were known, three defects of the plan's weighting came to light. First, with $b = 1$ on raw base units $V(z)$ is within 10^{-4} of one for 99.52% of the

TABLE 2. Order of Commitments, the Record of Each, and What Was Known When It Was Made (2026, Times in UTC)

When	Commitment or event	Record	Already known
9 Oct, 13:41:27	Analysis plan and configuration committed: cohort rule, scores (C-PR at $\lambda \in \{0.25, 0.5, 0.75\}$ and its end points, S-PR), W0 labels, statistics and rule A; amounts in raw base units, every ERC-20 token	`43d8045`	No log of this study extracted
9 Oct, 13:41:37–13:43:54	Scan A of the observation window; the BigQuery job was created 10 s after the plan commit	Job metadata; cost ledger	—
9 Oct, before 16:49	Cohort of 27,844 contracts, ego extraction, same-window and W0 evaluation; rule A judged	Summaries committed in `0d3a057`	Logs of the observation window
9 Oct, 16:49:32	Registration of W1: rule B with $\delta = 0.01$, F4–F6, sensitivity analyses S1 and S2, the companion study's comparison on trader outcomes (its rules R1–R3 and their wording); configuration status set to registered	`0d3a057`	Every same-window and W0 result of the registered analysis; W0 profit labels of the traders (exploratory run)
9 Oct, 16:49:33	Extraction script checks the committed registration and writes its manifest (1 s after the commit): commit and both SHA-256 hashes	Manifest	—
9 Oct, 16:49:40–16:50:11	Scan B of July–September 2026; job created 8 s after the commit; manifest closed at 16:50:55	Job metadata; manifest	No July–September log extracted for this study before this
9 Oct, 18:00:55	Registered W1 evaluation, then robustness and Sybil model, committed	`8d8b568`	W1 labels
9 Oct, 23:24:39	Registered benchmark and data description (BigQuery job times, amount shares) committed	`f33a87a`	W1 labels
Later	Revised analysis, a deviation from the plan and the registration: amounts in US dollars on five tokens (WETH, USDC, USDT0, ARB, WBTC), slopes $b = \ln 3/m$, a cap U on the C-PR authorization layer, and scores from the twelve months before each freeze with one cohort per freeze; constants $m_T = 170$, $m_A = 100$ and $U = 19{,}000$ dollars; rules A, B and R1–R3 applied with their thresholds; results committed	Commits in Table 3; results in `950a7a4`	Every registered result
Later	Exploratory checks of Appendix A	This paper's supplement	Every result above

Commit times are committer times, recorded in UTC+9 and converted. Job times are the creation and end times that BigQuery reports for each job. Full commit identifiers and file hashes are in Table 3.

transfers and 99.78% of the latest allowances, so the amount played no part in EndorseRank or AWP. Second, each layer of C-PR summed raw base units over tokens, although a USDC has 10^6 base units and a wrapped ether 10^{18}, and an unlimited allowance ($2^{256}-1$ base units, 45.3% of the latest allowances) absorbed nearly all of the row of any owner holding one. Third, the logs of every ERC-20 contract entered the graphs, although a token contract can write logs that name any address as an owner or a sender whether or not that address acted.

The revised analysis, defined with these results in view, values every amount in US dollars at the daily close of the DefiLlama price series [44], keeps the five tokens with the most rows in the twelve months before 30 June 2026 among those priced by a Chainlink market-price feed of low or medium market risk [45] (WETH, USDC, USDT0, ARB and WBTC), and computes every score from the twelve months before its freeze, with one cohort per freeze. With one year per freeze, no event older than a year enters a score, a same-window check or the history behind a label, and both freezes are scored on windows of equal length. It sets $b = \ln 3/m$ for EndorseRank and AWP, with the medians taken in the twelve months up to the W0 freeze so that no later event enters them, and caps each allowance of the C-PR authorization layer at $U = 19{,}000$ dollars, the 99th percentile of single transfers of the five tokens in the same months (18,575 dollars) rounded to two significant digits. A percentile of the finite allowances would not serve as the cap: their 99th percentile lies near 10^{63} dollars, so unlimited allowances would still take the rows of the owners holding them. All cohorts, graphs, proxies and spender labels are computed from the five tokens and the twelve-month windows, using the logs already extracted. Rules A and B are applied with their thresholds, δ and λ. The revised analysis, with its dollar amounts, five tokens and twelve-month score windows, is a deviation from the plan and the registration, decided with the registered results in view; Table 3 lists the commits that define it, and every verdict on its scores is a re-analysis.

E. WHAT WAS NOT PRE-SPECIFIED

Every contrast outside A1, A2, F1–F6, S1 and S2 was computed after the labels of its window were known: the comparison of C-PR with the payment walk on new approval pairs in W0, the S-PR contrasts, every contrast against a raw degree, and every comparison of C-PR with EndorseRank or AWP outside S1. The W0 profit labels of the traders had been computed in an exploratory run, and their coefficients were committed with the registration; the liquidation label of W0 had not been computed for any score of this study. The plan commit holds the plan, the configuration, the query that copies the logs and the BigQuery runner. The queries and scripts that build the cohort, the ego network, the pair tables and the W0 labels, the implementations of every score and of the bootstrap, and the drivers of the same-window and W0 evaluations entered the repository with the registration commit, after they had run on the observation window and after Rule A had been judged, so their text before that point is not on record; the unit tests of Section III entered with them. The W1 extraction and evaluation used them as committed.

TABLE 3. Commit Identifiers of the Pre-Specification Record (Times in UTC) and of the Revised Analysis, and SHA-256 Hashes of the Registration Files and the Plan

Plan and configuration (rule A), 9 Oct 13:41:27
`43d80457142dc002c4410d4af1c3c164f39a3272`
Registration of W1 (rule B, R1–R3), 9 Oct 16:49:32
`0d3a057e4f0c8473d36d89fbe33c24deff0c9911`
Registered results committed, 9 Oct 18:00:55
`8d8b568ecaf882cccc6caef60b444823966f5ce8`
Registered benchmark and data description, 9 Oct 23:24:39
`f33a87a92f3f894c829451e55cc7ff9a33d74e3f`
Revised analysis (US-dollar amounts, five tokens, scores from the twelve months before each freeze), defined after the registered results were known
US-dollar amounts, token rule and slopes
`a8974759f8a8e6e0c8b07c23481ff429eaa492ee`
Five tokens and price inputs
`09dfc323de2c3f0ffa40c7a6a0c5adf0b94c1122`
Cap rule of the C-PR authorization layer
`2705b28a4653d47a6940cca064bf923562db8f9d`
Score windows: the twelve months before each freeze
`e6c352dcbc5ba14e7bcd25df7bdfeefedb1091ce`
Cohorts and constants; code revision of the results
`0a9421563a529762662ccaef4d86ac96bfad03f6`
Revised results committed
`950a7a4cd9401002b3be9348797b4a8dd9d9f075`
SHA-256 of the registration document, CRLF line ends (recorded in the manifest)
`2126c8dc3fe3c0f9273aa4ecfc1e15bd`
`dc1f5a6f65e98548b9528f2b18ca2a6b`
SHA-256 of the registration document, LF line ends (as git stores it)
`34cd6dfeb56b8fb8a553cb24afbff06c`
`6a94e6f69a2eb7f7d642b0cc39e54b2c`
SHA-256 of the configuration, CRLF line ends (recorded in the manifest)
`d6a30adbcabcf511d6892332da73639e`
`fdbece25891b392e13e90faf09b7a833`
SHA-256 of the configuration, LF line ends (as git stores it)
`2ea235021241fa408f8776d55c2b594d`
`34ced0ca1dad44e16f22775c5db8297f`
SHA-256 of the plan in `43d8045`, LF line ends
`7ec80d49c17331bd10e086fecd909a3f`
`36020dbb284f0dbfbb81a8549f84fd9f`

Each SHA-256 value is printed over two lines; join them without a space.

Every check in Appendix A was run after all of the above.

V. DATA AND MEASUREMENT

A. SOURCES AND COHORT

All logs come from Arbitrum One (chain ID 42161), an optimistic rollup on Ethereum [46], [47], through the public BigQuery table

`bigquery-public-data.goog_blockchain_arbitrum_one_us.logs`.

Scan A copied the logs of the observation window, 1 October 2023 to 30 June 2026, and scan B those of July to September 2026. In the revised analysis each score uses the year that ends at its freeze: July 2025 through June 2026 for W1 and the same-window checks, April 2025 through March 2026 for W0. The registered analysis scores every event of the observation window up to its freeze. Ages are measured back from the freeze. Approval and transfer logs are selected by their ERC-20 event signatures and exactly three topics, which leaves out ERC-721 tokens; zero-value transfers are dropped, zero-value approvals are kept because they delete an allowance. A close is an `EventLog1` record named `PositionDecrease` from the GMX V2 event emitter, which counts partial closes and liquidations alike; it counts as a liquidation if its order type is liquidation or if the liquidation handler emitted it [16].

The matched cohort holds every contract that received non-zero approvals from at least three distinct owners in the score window. In the revised analysis the rule counts the approvals of the five tokens in the twelve months before each freeze: 6,483 contracts at the freeze of 30 June 2026, the matched cohort, and 6,343 at the W0 freeze, 5,383 of them in both. The plan's cohort counts the approvals of every token in the observation window: of the 29,022 spender addresses that met the rule, 27,844 held contract code at block 513,216,506 (read with `eth_getCode`), while 1,121 held none and 57 held an EIP-7702 delegation designator [48]; those two groups are externally owned accounts and were left out. Both revised cohorts lie inside the plan's cohort. Only events with a cohort contract at one end are extracted: an approval in which a cohort contract is owner or spender, and a transfer in which it is sender or recipient. Each graph is therefore the one-hop ego network of the cohort, in the revised analysis of the cohort of its freeze. In the twelve months before 30 June 2026 the revised ego tables hold 85,986,562 approval and 442,722,039 transfer logs, 63.3% and 76.2% of the logs that the plan's ego tables hold in the same months; over the observation window the plan's ego tables hold 389,866,882 and 1,538,404,067.

B. GRAPHS

At the freeze of 30 June 2026 the authorization graph has 2,514,846 nodes and 3,808,949 edges in the revised analysis and 13,055,150 nodes and 22,321,470 edges in the registered one; the payment graph has 3,603,300 nodes and 8,426,124 edges, and 23,029,298 nodes and 59,737,617 edges. In the revised analysis these node counts include the end points of some pairs that carry only dust amounts, whose bounded weight V evaluates to zero in floating point; the edge counts leave such pairs out, and so do the solvers. EndorseRank's graph therefore has 2,513,982 nodes and AWP's 3,601,727, while the amount-weighted layers of C-PR and S-PR keep those pairs, so their payment layer has 8,432,635 edges (Table 1). In the registered analysis no weight is zero and the counts coincide. Of the latest positive allowances, 49.8% are unlimited in the revised analysis, where 55.7% reach the cap U, and 45.3% in the registered one. Cohort contracts absent from a walk's graph score zero under that walk (S2 changes this): in the revised analysis 191 are outside the authorization graph, 1,192 outside the payment graph and 30 outside both.

C. COHORTS AND LABELS

The spender cohort of a window holds every contract that has a positive latest allowance at the freeze in the approval data of the score window; it includes contracts outside the matched cohort that a cohort contract approved. Their first label counts new approval pairs: owners that gave the spender a positive approval during the label window although the pair held no positive allowance at the freeze. Their second label counts

new transfer senders: addresses other than the spender that sent it a transfer during the label window but had not done so between the start of the score window and the freeze. In W0 the spender cohort holds 7,914 contracts in the revised and 33,101 in the registered analysis, and in W1 7,960 and 34,587; Table 4 gives the number that gained each label.

The W1 trader cohort consists of contract accounts on GMX V2 that appear in either freeze-date graph and closed three or more positions between July and September 2026: 70 contracts in both analyses, 9 of them with at least one liquidation and 1 with every close liquidated. Their label is the liquidation-free close rate, one minus the share of their closes in the label window that ended in liquidation. It is built from neither layer, and as a rate it does not reward trading more. Contract accounts closed 133,685 positions in the observation window (6,473 through liquidation) and 6,805 in the third quarter of 2026 (182 through liquidation).

D. STATISTICS

Every coefficient is the tie-corrected Kendall τ_b [15], computed with SciPy, with a 95% percentile interval from a paired bootstrap over the contracts of the cohort [49], [50]. Resample indices are drawn with NumPy's default generator from seed 42, one index matrix per cohort, and the same matrix serves every score on a label, so each difference between two scores is a paired difference. Every result uses 400 resamples. The supplementary script recomputes every decision contrast except those of S2 from the published per-contract score files and reproduces the published values exactly (Appendix A).

VI. RESULTS

Each subsection reports the revised analysis first, a reanalysis under the rules, and then the registered analysis, the test that the rules were fixed for; Tables 4 and 5 and Figs. 2 and 3 give both.

A. SAME WINDOW: RULE A'S SECONDARY PART FAILS

On the matched cohort, C-PR at $\lambda = 0.5$ lies between its two layers on every family. In the revised analysis it reaches 0.400 $[0.385, 0.416]$ on the transfer family, far below the payment walk's interval $[0.691, 0.709]$; 0.449 $[0.437, 0.463]$ on the authorization family, below the authorization walk's interval $[0.550, 0.572]$; and 0.142 $[0.128, 0.156]$ on the Sybil-stability family, above the authorization walk (0.028 $[0.014, 0.042]$) but below the payment walk (0.402 $[0.389, 0.417]$). The registered scores give the same pattern: 0.296 $[0.289, 0.303]$ against $[0.522, 0.534]$, 0.418 $[0.412, 0.423]$ against $[0.506, 0.517]$, and 0.129 $[0.124, 0.134]$ between 0.020 $[0.015, 0.026]$ and 0.311 $[0.304, 0.317]$. All three parts of the secondary criterion therefore fail in both analyses. The paired contrasts agree: C-PR minus the payment walk on the transfer family is -0.300 $[-0.318, -0.281]$ in the revised and -0.232 $[-0.239, -0.224]$ in the registered analysis, and C-PR minus the authorization walk on the authorization family -0.112 $[-0.124, -0.100]$ and -0.094 $[-0.099, -0.088]$. These checks come from the events that build the graphs, so they show which construct a score tracks and are not external evidence [42].

B. W0: RULE A'S PRIMARY PART FAILS

Table 4 gives every score against the later labels, Table 5 the contrasts, and Fig. 2 the contrasts that the rules fixed. In the revised analysis A1 holds: C-PR is not worse than the authorization walk on new approval pairs ($+0.031$ $[0.021, 0.042]$). A2 fails: on new transfer senders C-PR lies below the payment walk (-0.030 $[-0.040, -0.020]$). With its primary part failed, Rule A answers in the negative whether C-PR beats a walk over either layer alone. The registered analysis, the test that Rule A was written for, reaches the same verdicts, with A1 $+0.008$ $[0.003, 0.013]$ and A2 -0.030 $[-0.034, -0.026]$.

A comparison that Rule A did not contain pointed the other way. In the registered analysis, the only one that existed when Rule B was written, C-PR ranked new approval pairs much better than the payment walk alone (0.236 against 0.142; $+0.094$ $[0.085, 0.103]$). Because this contrast was examined after the labels were known, we treated it as a hypothesis and wrote Rule B to test it on a window that this study had not yet extracted. The revised scores give $+0.151$ $[0.130, 0.170]$.

C. W1: RULE B IS NOT MET

In the revised analysis F1 holds: C-PR ranks the spenders that later gain approval pairs at $\tau = 0.316$ $[0.300, 0.330]$ against 0.146 $[0.126, 0.168]$ for the payment walk, a difference of $+0.170$ $[0.149, 0.189]$, so the W0 gain held again in a window that this study extracted after the rule was fixed. F2 fails: on new transfer senders C-PR reaches 0.324 $[0.308, 0.337]$ against 0.348 $[0.335, 0.363]$, a difference of -0.025 $[-0.038, -0.015]$ whose whole interval lies beyond the margin of -0.01, so the data show that C-PR ranks later payers worse than the payment walk by more than the registration allowed. F3 fails narrowly: on the traders' liquidation-free close rate C-PR reaches 0.282 $[0.162, 0.397]$ and the payment walk 0.282 $[0.162, 0.394]$, and the interval of their difference, $+0.000$ $[-0.010, 0.010]$, has its lower end just below -0.01. Because the rule required all three parts, it is not met, and we do not claim that C-PR improves on the payment walk.

The registered analysis, the test that the registration fixed, reaches the same verdicts (Table 5). F1 holds ($+0.101$ $[0.092, 0.111]$; 0.237 against 0.136) and F2 fails with its whole interval beyond the margin (-0.025 $[-0.029, -0.021]$; 0.235 against 0.260). F3 fails there too: C-PR reaches 0.095 $[-0.102, 0.274]$ against 0.138 $[-0.048, 0.310]$ for the payment walk, a difference of -0.044 $[-0.100, -0.002]$ whose interval lies below zero at the registered seed and reaches zero under some other seeds (Appendix A). The weakness that the registration anticipated for F3 did occur in this analysis: both walk coefficients have intervals that include zero, while the authorization walk reaches 0.310 $[0.181, 0.432]$ and the in-approve degree 0.394 $[0.261, 0.524]$ on the same label. The two analyses therefore differ on F3 in kind. In the registered

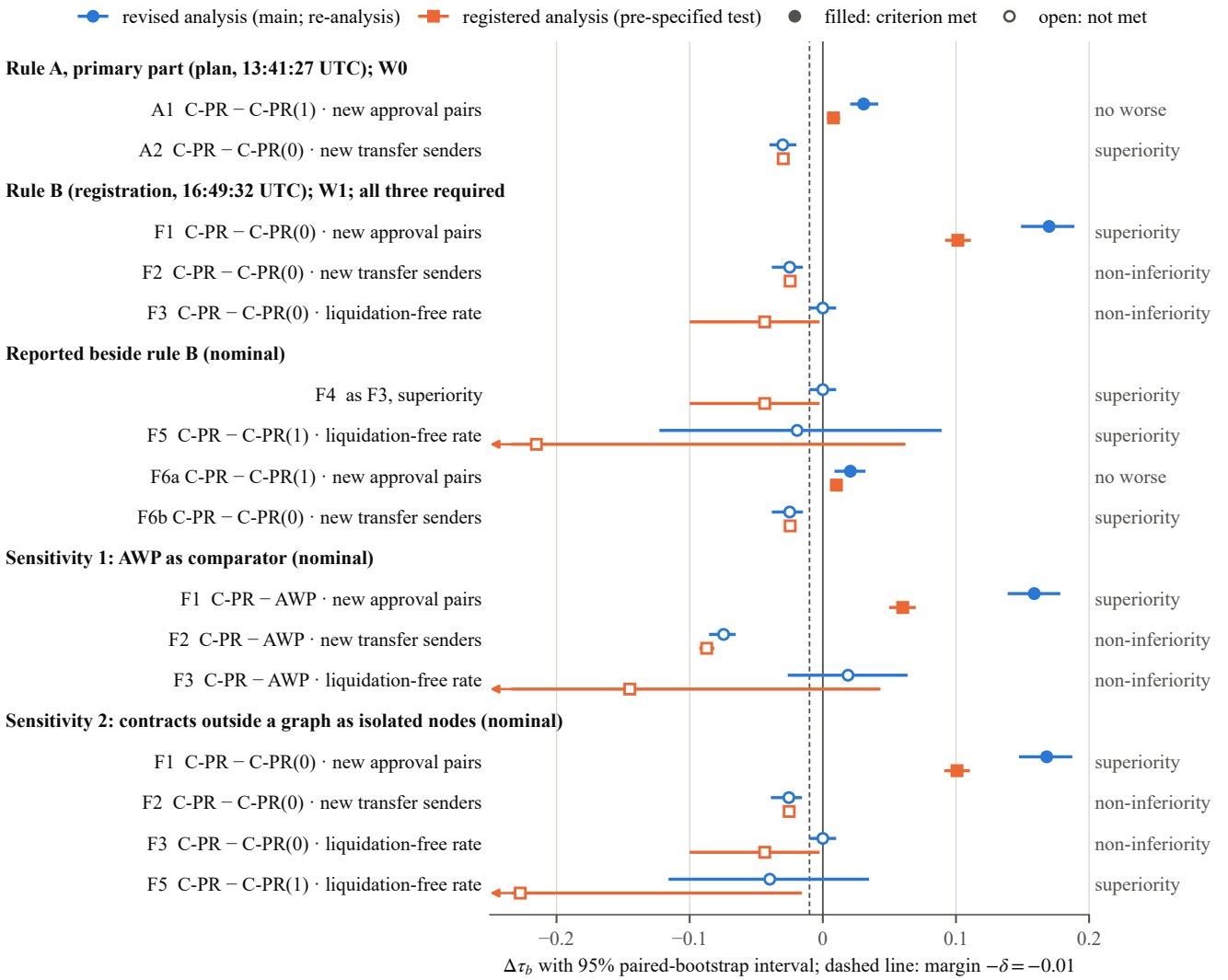


FIGURE 2. **Contrasts fixed by Rules A and B, the results registered to be reported beside Rule B, and the two registered sensitivity analyses, with 95% paired-bootstrap intervals (400 resamples, seed 42). Circles: revised analysis (US dollars, five tokens, scores from the twelve months before each freeze), the main analysis, defined after the registered results were known and a re-analysis under the same rules; squares: registered analysis, the pre-specified test. Filled markers: criterion met; open markers: not met. The dashed line is the margin $-\delta = -0.01$; arrows mark interval ends beyond the plotted range. Only A1–A2 and F1–F3 decide a rule; the other verdicts are nominal. S1's F4 has the contrast of its F3 and is omitted. Values in Table 5.**

analysis the raw-amount payment walk and C-PR both ranked the traders weakly and C-PR had the lower coefficient; in the revised analysis every score of Table 4 ranked them with an interval above zero and C-PR was level with the payment walk, failing only on precision. The highest coefficient on this label in both analyses belongs to the in-approve degree, and in a contrast computed afterwards from the revised score files C-PR ranks the traders below it (-0.112 $[-0.194, -0.026]$).

D. RESULTS REPORTED BESIDE THE DECISION

These verdicts are nominal: each uses an unadjusted 95% interval, and none was conditioned on another. In both analyses F4 fails, since C-PR is not ahead of the payment walk on the trader label; F5 fails, since C-PR and the authorization walk cannot be separated on that label (-0.019 $[-0.123, 0.089]$ revised; -0.215 $[-0.472, 0.062]$ registered); and F6, Rule A's primary part rerun on W1, fails as in W0: C-PR is no worse than the authorization walk on new approval pairs ($+0.021$ $[0.009, 0.032]$; $+0.010$ $[0.005, 0.014]$) but lies below the payment walk on new senders.

Neither sensitivity analysis changes the decision in either analysis. With AWP as the comparator (S1), F1 holds ($+0.159$ $[0.139, 0.178]$ revised; $+0.060$ $[0.050, 0.070]$ registered) and F2 fails with its whole interval far below the margin: AWP ranks new senders at 0.398 $[0.386, 0.412]$ in the revised and 0.322 $[0.316, 0.329]$ in the registered analysis, and C-PR's deficit is -0.075 $[-0.085, -0.065]$ and -0.087 $[-0.093, -0.082]$. Against the published payment-based method, then, C-PR trades a measurable part of its ranking of later inflow for its gain on later approvals. F3

TABLE 4. Kendall τ_b Between Scores at the Freeze and Later Labels, With 95% Paired-Bootstrap Intervals

Revised analysis, the main analysis, a re-analysis under the pre-specified rules (US dollars, five tokens, twelve-month score windows)					
	W0 spenders ($n = 7{,}914$)		W1 spenders ($n = 7{,}960$)		W1 traders ($n = 70$)
Score	New approval pairs	New transfer senders	New approval pairs	New transfer senders	Liquidation-free close rate
In-approve degree at the freeze	0.392 [0.377, 0.406]	0.284 [0.267, 0.299]	0.381 [0.365, 0.396]	0.269 [0.250, 0.285]	0.394 [0.261, 0.524]
Transfer in-degree at the freeze	0.203 [0.183, 0.224]	0.447 [0.435, 0.459]	0.165 [0.143, 0.186]	0.439 [0.426, 0.454]	0.253 [0.116, 0.363]
EndorseRank	0.303 [0.287, 0.319]	0.272 [0.257, 0.286]	0.299 [0.284, 0.314]	0.259 [0.243, 0.275]	0.297 [0.174, 0.418]
AWP	0.196 [0.178, 0.213]	0.409 [0.396, 0.421]	0.157 [0.137, 0.178]	0.398 [0.386, 0.412]	0.263 [0.140, 0.379]
C-PR ($\lambda = 1$), authorization walk	0.298 [0.282, 0.313]	0.267 [0.250, 0.282]	0.295 [0.280, 0.310]	0.257 [0.241, 0.274]	0.302 [0.177, 0.425]
C-PR ($\lambda = 0.75$)	0.327 [0.311, 0.342]	0.327 [0.313, 0.339]	0.316 [0.298, 0.330]	0.315 [0.299, 0.329]	0.284 [0.160, 0.398]
C-PR ($\lambda = 0.5$), primary	0.329 [0.312, 0.343]	0.336 [0.322, 0.348]	0.316 [0.300, 0.330]	0.324 [0.308, 0.337]	0.282 [0.162, 0.397]
C-PR ($\lambda = 0.25$)	0.329 [0.312, 0.343]	0.345 [0.330, 0.357]	0.315 [0.299, 0.328]	0.332 [0.317, 0.346]	0.282 [0.162, 0.397]
C-PR ($\lambda = 0$), payment walk	0.178 [0.160, 0.197]	0.366 [0.351, 0.378]	0.146 [0.126, 0.168]	0.348 [0.335, 0.363]	0.282 [0.162, 0.394]
S-PR	0.190 [0.171, 0.209]	0.377 [0.363, 0.389]	0.162 [0.140, 0.184]	0.367 [0.354, 0.381]	0.300 [0.175, 0.411]
Registered analysis, the pre-specified test (raw base units, every ERC-20 token)					
	W0 spenders ($n = 33{,}101$)		W1 spenders ($n = 34{,}587$)		W1 traders ($n = 70$)
Score	New approval pairs	New transfer senders	New approval pairs	New transfer senders	Liquidation-free close rate
In-approve degree at the freeze	0.252 [0.244, 0.260]	0.156 [0.148, 0.166]	0.262 [0.255, 0.269]	0.179 [0.171, 0.186]	0.394 [0.261, 0.524]
Transfer in-degree at the freeze	0.148 [0.138, 0.158]	0.302 [0.294, 0.310]	0.145 [0.135, 0.155]	0.303 [0.296, 0.309]	0.331 [0.209, 0.441]
EndorseRank	0.241 [0.234, 0.248]	0.226 [0.219, 0.234]	0.233 [0.226, 0.240]	0.210 [0.203, 0.217]	0.306 [0.176, 0.421]
AWP	0.196 [0.187, 0.206]	0.340 [0.333, 0.348]	0.177 [0.166, 0.187]	0.322 [0.316, 0.329]	0.240 [0.090, 0.362]
C-PR ($\lambda = 1$), authorization walk	0.228 [0.219, 0.236]	0.122 [0.113, 0.133]	0.227 [0.219, 0.234]	0.133 [0.125, 0.142]	0.310 [0.181, 0.432]
C-PR ($\lambda = 0.75$)	0.236 [0.229, 0.243]	0.217 [0.209, 0.226]	0.237 [0.230, 0.244]	0.228 [0.220, 0.236]	0.088 [−0.110, 0.269]
C-PR ($\lambda = 0.5$), primary	0.236 [0.229, 0.243]	0.224 [0.216, 0.234]	0.237 [0.230, 0.244]	0.235 [0.227, 0.243]	0.095 [−0.102, 0.274]
C-PR ($\lambda = 0.25$)	0.234 [0.227, 0.241]	0.230 [0.222, 0.240]	0.235 [0.228, 0.242]	0.240 [0.232, 0.247]	0.101 [−0.096, 0.280]
C-PR ($\lambda = 0$), payment walk	0.142 [0.133, 0.151]	0.254 [0.246, 0.262]	0.136 [0.125, 0.145]	0.260 [0.252, 0.266]	0.138 [−0.048, 0.310]
S-PR	0.177 [0.167, 0.186]	0.245 [0.236, 0.253]	0.169 [0.159, 0.178]	0.251 [0.244, 0.258]	0.312 [0.183, 0.422]

W0: scores frozen on 31 March 2026, labels counted from April to June 2026. W1: scores frozen on 30 June 2026, labels counted from July to September 2026. In the revised analysis every score is computed from the twelve months before its freeze, in the registered analysis from every extracted event since 1 October 2023. Spenders are the contracts that hold a positive latest allowance at the freeze; traders are the GMX V2 accounts that hold contract code, are nodes of the freeze-date graph and close at least three positions in the label window. In the revised analysis 1,732 W0 spenders gained a new approval pair and 1,422 a new transfer sender, and 1,495 and 1,371 W1 spenders did; in the registered analysis 3,403, 3,042, 2,850 and 2,812. Of the 70 W1 traders, 9 had at least one liquidation. The degree, EndorseRank and AWP rows are reference scores; the companion study [18] reports their comparisons in full. Intervals: 400 paired resamples over the contracts of the cohort, seed 42.

under S1 fails as well ($+0.019$ $[-0.026, 0.064]$; -0.145 $[-0.348, 0.043]$); on 70 traders the revised interval is 0.090 wide, so it leaves a loss of 0.026 possible but does not show one. With absent contracts kept as isolated nodes (S2), no verdict changes and the F1 to F3 point estimates move by 0.002 at most. F5 moves more, because the isolated-node rule lifts the traders outside the authorization graph from zero to the score of a node without in-edges and so changes the ties of the authorization walk: in the revised analysis the authorization walk's coefficient on the trader label rises from 0.302 to 0.322 and F5 stays unresolved (-0.040 $[-0.116, 0.035]$), and in the registered analysis F5 becomes a resolved deficit (-0.227 $[-0.426, -0.016]$).

E. THE WEIGHT SWEEP

Fig. 3 plots each label against λ. On new approval pairs the large step comes at the lower end: switching the authorization layer on lifts C-PR from 0.146 to 0.315 in the revised W1 and from 0.136 to 0.235 in the registered W1, and the interior values differ by less than 0.003. On new transfer senders the interior curve falls slowly as λ rises (0.332, 0.324, 0.315 in the revised W1; 0.240, 0.235, 0.228 in the registered one), and every interior value lies below the payment walk. At the upper end the authorization walk drops on new senders (0.257 and 0.133), because at $\lambda = 1$ the payment rows of contracts without authorization out-edges disappear. The F1 gain over the payment walk is thus a property of switching the authorization layer on, not of the value $\lambda = 0.5$, and no interior value recovers the payment walk's ranking of later payers. On the trader label the revised curve is flat at 0.282 to 0.284 across the interior, level with the payment walk (0.282) and a little below the authorization walk (0.302), whereas the registered curve stays between 0.088 and 0.101, below a payment walk at 0.138 and an authorization walk at 0.310. Appendix A gives the paired steps from $\lambda = 0$ to 0.25, from 0.25 to 0.75 and from 0.75 to 1, and the rank agreement between these values.

F. SEEDED RESTARTS

S-PR stays close to the payment walk on new senders and far below the authorization walk on new approval pairs. In W0, S-PR minus the authorization walk on new approval pairs is -0.108 $[-0.128, -0.088]$ in the revised and -0.051 $[-0.060, -0.043]$ in the registered analysis, and S-PR minus the payment walk on new senders is $+0.011$ $[0.004, 0.018]$ and -0.009 $[-0.012, -0.006]$. Its restarts lift its coefficient on new approval pairs only part of the way from the payment walk toward the authorization walk (0.190 against 0.178 and 0.298 in the revised W0; 0.177 against 0.142 and 0.228 in the registered one; no paired contrast with the payment walk was

TABLE 5. Paired Contrasts in Both Analyses: The Pre-Specified Rules, the Registered Reports and Sensitivity Analyses, and Contrasts Computed After the Labels Were Known

		Revised analysis (re-analysis)		Registered analysis (the test)	
Contrast (a minus b) and label	Test	$\Delta\tau$ [95% CI]	Verdict	$\Delta\tau$ [95% CI]	Verdict
Rule A, fixed in the plan (43d8045) before any log of this study was extracted; W0 spenders					
A1: C-PR − C-PR ($\lambda = 1$), new approval pairs	no worse	+0.031 [0.021, 0.042]	holds	+0.008 [0.003, 0.013]	holds
A2: C-PR − C-PR ($\lambda = 0$), new transfer senders	superiority	−0.030 [−0.040, −0.020]	fails	−0.030 [−0.034, −0.026]	fails
Rule A, primary part (A1 and A2)			**not met**		**not met**
Rule A, secondary part (same window)			**not met**		**not met**
Rule B, registered (0d3a057) before this study extracted the W1 logs; F1, F2 and F3 all required; margin $\delta = 0.01$					
F1: C-PR − C-PR ($\lambda = 0$), new approval pairs	superiority	+0.170 [0.149, 0.189]	holds	+0.101 [0.092, 0.111]	holds
F2: C-PR − C-PR ($\lambda = 0$), new transfer senders	non-inferiority	−0.025 [−0.038, −0.015]	fails	−0.025 [−0.029, −0.021]	fails
F3: C-PR − C-PR ($\lambda = 0$), liquidation-free rate	non-inferiority	+0.000 [−0.010, 0.010]	fails	−0.044 [−0.100, −0.002]	fails
Rule B (F1, F2 and F3)			**not met**		**not met**
Registered to be reported in a fixed order beside the decision (nominal)					
F4: as F3	superiority	+0.000 [−0.010, 0.010]	fails	−0.044 [−0.100, −0.002]	fails
F5: C-PR − C-PR ($\lambda = 1$), liquidation-free rate	superiority	−0.019 [−0.123, 0.089]	fails	−0.215 [−0.472, 0.062]	fails
F6a: C-PR − C-PR ($\lambda = 1$), new approval pairs	no worse	+0.021 [0.009, 0.032]	holds	+0.010 [0.005, 0.014]	holds
F6b: C-PR − C-PR ($\lambda = 0$), new transfer senders	superiority	−0.025 [−0.038, −0.015]	fails	−0.025 [−0.029, −0.021]	fails
Registered sensitivity analysis S1: AWP as the comparator (nominal)					
F1: C-PR − AWP, new approval pairs	superiority	+0.159 [0.139, 0.178]	holds	+0.060 [0.050, 0.070]	holds
F2: C-PR − AWP, new transfer senders	non-inferiority	−0.075 [−0.085, −0.065]	fails	−0.087 [−0.093, −0.082]	fails
F3: C-PR − AWP, liquidation-free rate	non-inferiority	+0.019 [−0.026, 0.064]	fails	−0.145 [−0.348, 0.043]	fails
Registered sensitivity analysis S2: contracts outside a graph kept as isolated nodes (nominal)					
F1: C-PR − C-PR ($\lambda = 0$), new approval pairs	superiority	+0.168 [0.147, 0.187]	holds	+0.101 [0.091, 0.111]	holds
F2: C-PR − C-PR ($\lambda = 0$), new transfer senders	non-inferiority	−0.025 [−0.039, −0.016]	fails	−0.025 [−0.029, −0.022]	fails
F3: C-PR − C-PR ($\lambda = 0$), liquidation-free rate	non-inferiority	+0.000 [−0.010, 0.010]	fails	−0.044 [−0.100, −0.002]	fails
F5: C-PR − C-PR ($\lambda = 1$), liquidation-free rate	superiority	−0.040 [−0.116, 0.035]	fails	−0.227 [−0.426, −0.016]	fails
S1, decision (F1, F2 and F3)			**not met**		**not met**
S2, decision (F1, F2 and F3)			**not met**		**not met**
Computed after the labels of the window were known; no rule					
W0: C-PR − C-PR ($\lambda = 0$), new approval pairs	—	+0.151 [0.130, 0.170]		+0.094 [0.085, 0.103]	
W0: S-PR − C-PR ($\lambda = 1$), new approval pairs	—	−0.108 [−0.128, −0.088]		−0.051 [−0.060, −0.043]	
W0: S-PR − C-PR ($\lambda = 0$), new transfer senders	—	+0.011 [0.004, 0.018]		−0.009 [−0.012, −0.006]	
W0: C-PR − AWP, new transfer senders	—	−0.073 [−0.083, −0.063]		−0.116 [−0.122, −0.109]	
W0: C-PR ($\lambda = 1$) − in-approve degree, new approval pairs	—	−0.094 [−0.104, −0.083]		−0.025 [−0.030, −0.019]	
W0: EndorseRank − in-approve degree, new approval pairs	—	−0.090 [−0.100, −0.079]		−0.011 [−0.016, −0.006]	
W0: AWP − transfer in-degree, new transfer senders	—	−0.038 [−0.043, −0.032]		+0.038 [0.035, 0.042]	
W1: EndorseRank − in-approve degree, new approval pairs	—	−0.082 [−0.090, −0.073]		−0.029 [−0.032, −0.025]	
W1: AWP − transfer in-degree, new transfer senders	—	−0.041 [−0.047, −0.034]		+0.020 [0.016, 0.023]	
W1: C-PR − EndorseRank, new approval pairs	—	+0.017 [0.006, 0.029]		+0.004 [−0.001, 0.008]	

$\Delta\tau = \tau_a - \tau_b$ on the same resamples. C-PR without a λ is the primary setting $\lambda = 0.5$; liquidation-free rate: the liquidation-free close rate of the W1 traders. Superiority holds when the interval lies above zero, non-inferiority when its lower end lies above $-\delta = -0.01$, and "no worse" when the interval contains zero or lies above it. The revised column, the main analysis, applies the rules and thresholds of the plan and the registration to scores defined after the registered results were known (amounts in US dollars, five tokens, scores from the twelve months before each freeze), so none of its verdicts is a registered result; the registered column is the test that the plan and the registration fixed. Nominal verdicts use unadjusted 95% intervals; the fixed order was an order of reporting, not a gatekeeping sequence. The degree, EndorseRank and AWP rows are reference contrasts reported in full in [18]. Intervals: 400 paired resamples, seed 42.

computed). With $d = 0.85$ only 15% of the mass, plus the dangling mass, passes through the restart vector. Authorization edges thus help the walk on later approvals when they carry it, fully in the authorization walk and mixed in C-PR, and help little when they only decide where it restarts. On later payers the order reverses: in contrasts computed afterwards from the revised score files, C-PR ranks new approval pairs above S-PR in both windows ($+0.139$ $[0.121, 0.156]$ and $+0.154$ $[0.134, 0.174]$) but new senders below it (-0.041 $[-0.049, -0.032]$ and -0.044 $[-0.052, -0.037]$).

G. WALKS AGAINST COUNTS

No walk ranked new approval pairs as well as the raw in-approve degree, by point estimate in either analysis and by every paired contrast that was computed. In W0 the authorization walk falls short of the degree by -0.094 $[-0.104, -0.083]$ in the revised and -0.025 $[-0.030, -0.019]$ in the registered analysis, and EndorseRank by -0.090 $[-0.100, -0.079]$ and -0.011 $[-0.016, -0.006]$; in W1 EndorseRank falls short by -0.082 $[-0.090, -0.073]$ and -0.029 $[-0.032, -0.025]$, and the authorization walk's point estimates lie below the degree's (0.295 against 0.381 and 0.227 against 0.262), with no paired contrast computed. In paired contrasts computed afterwards from the revised score files, C-PR falls short of the degree in both windows (-0.063 $[-0.076, -0.052]$ and -0.065 $[-0.077, -0.054]$), and so does EndorseRank with AWP's activity restarts (-0.060 $[-0.070, -0.052]$ and

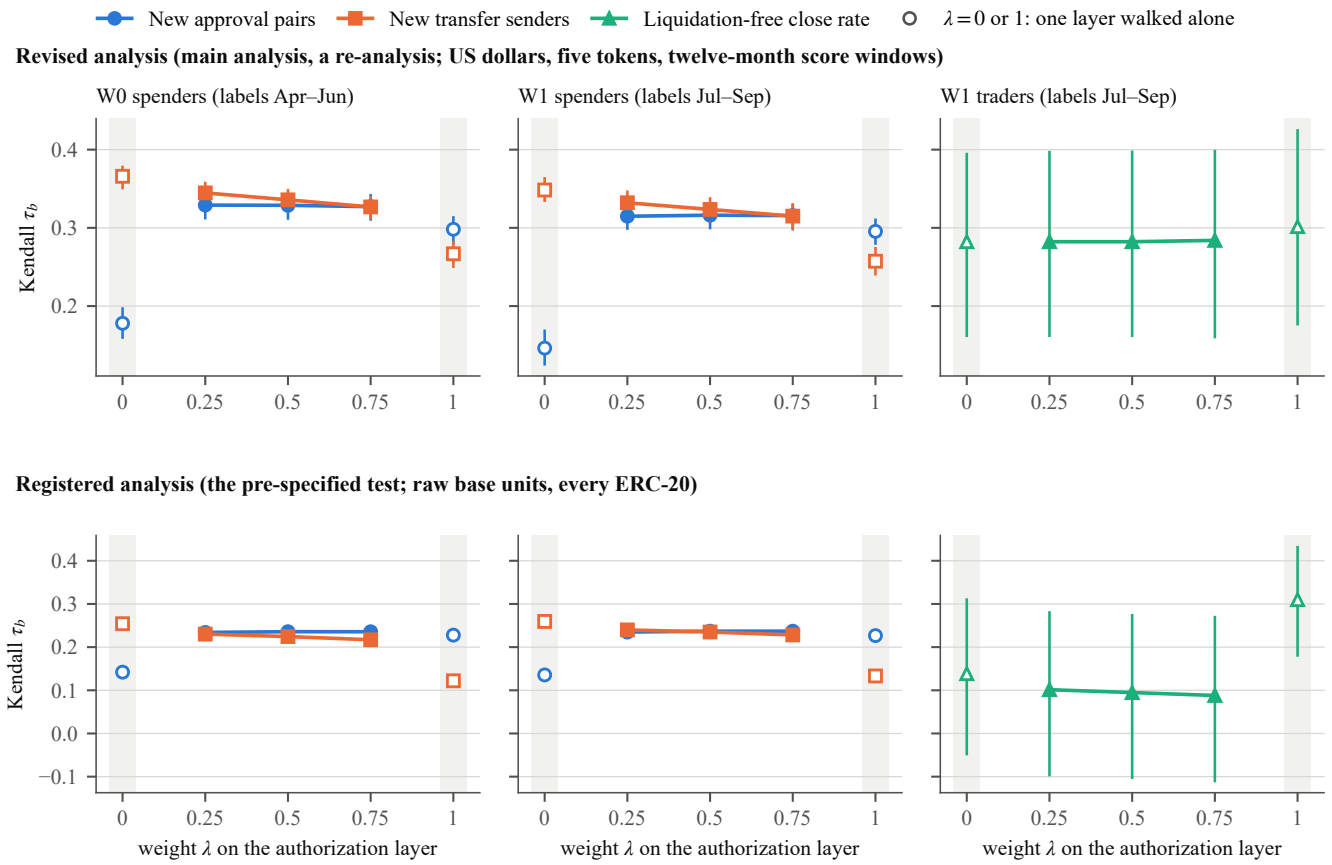


FIGURE 3. Kendall τ_b of C-PR with each later label against the authorization weight λ, with 95% paired-bootstrap intervals. Filled markers joined by lines: the pre-specified grid $\lambda \in \{0.25, 0.5, 0.75\}$; open markers in the shaded bands: $\lambda = 0$ and $\lambda = 1$, the payment and authorization walks, separate operators that the line does not join. Top row: revised analysis (the main analysis, a re-analysis); bottom row: registered analysis (the pre-specified test). Each row shares its vertical axis. Values in Table 4.

-0.046 $[-0.055, -0.038]$); in the registered analysis C-PR's point estimates on new approval pairs lie below the degree in both windows (Table 4), with no paired contrast computed. On new transfer senders the counts lead as well in the revised analysis. C-PR ranks them below the transfer in-degree in both windows (-0.111 $[-0.121, -0.100]$ and -0.115 $[-0.128, -0.105]$), and so does AWP (W0 -0.038 $[-0.043, -0.032]$; W1 -0.041 $[-0.047, -0.034]$), whereas in the registered analysis AWP ranks them above the in-degree ($+0.038$ $[0.035, 0.042]$ and $+0.020$ $[0.016, 0.023]$). The payment walk of C-PR, with uniform restarts and unbounded amounts, has lower point estimates than the in-degree in every case. The companion study reports the reference comparisons in full [18]; they are repeated here because they bound what any walk in this paper can claim. Proposition 2 accounts for the authorization side: for most spenders the walk is a reweighted owner count, and on this arrival label the reweighting did not help. The degree results agree with preferential attachment [29], [30] and with the strength of degree baselines in link prediction [31], [32].

H. AGAINST THE REFERENCE WALKS, AFTER THE LABELS

Rules A and B compare C-PR with its own layers, because only that contrast changes the coupling alone. Contrasts with the reference walks of the companion study [18], computed afterwards from the revised score files with the resamples of the rules, place it further; none enters a rule, and apart from two rows of Table 5 the registered analysis has no counterpart. On each spender label and in each window C-PR leads EndorseRank (new approval pairs $+0.026$ $[0.016, 0.037]$ and $+0.017$ $[0.006, 0.029]$; new senders $+0.064$ $[0.052, 0.075]$ and $+0.064$ $[0.051, 0.076]$). Against EndorseRank with AWP's activity restarts it is level on new approval pairs in W0 (-0.003 $[-0.014, 0.008]$) and below it in W1 (-0.019 $[-0.029, -0.008]$). Against AWP, C-PR's lead on new approval pairs (W0 $+0.133$ $[0.114, 0.153]$; W1 $+0.159$ $[0.139, 0.178]$, F1 of S1) comes largely from the spenders that AWP scores zero because they have no transfer edge: among the 6,489 W0 and 6,567 W1 spenders with an incoming transfer before the freeze, C-PR and AWP are level on new approval pairs in W0 ($+0.008$ $[-0.002, 0.020]$) and C-PR leads by 0.022 $[0.011, 0.033]$ in W1, while AWP ranks new senders higher by 0.036 $[0.026, 0.044]$ and 0.037 $[0.028, 0.046]$, as it does over all spenders (by 0.073

$[0.063, 0.083]$ in W0 and 0.075 $[0.065, 0.085]$ in W1). Out of window, what the authorization layer adds to a payment walk is thus mainly an order among the contracts that the payment walk leaves tied, with a small lead on later approvals among the others in W1, while on later payers C-PR trails AWP in every subset and window.

I. PRECISION AND THE MARGIN

The margin came from the precision of a spender contrast and was applied as it stood to the trader contrast. On the spenders it was attainable: F2's interval is 0.023 wide in the revised and 0.008 wide in the registered analysis, and F2 failed because its whole interval lay beyond the margin, not because the interval was wide. On the traders it lay at the edge of what 70 contracts allow: the F3 interval is 0.020 wide in the revised and 0.098 wide in the registered analysis. With the two coefficients equal, the revised F3 could hold only with an interval narrower than about 2δ; at the registered seed its half-width is 0.01004, and across 50 further bootstrap seeds the lower end ranges from -0.0146 to -0.0098 and clears the margin in one seed (Appendix A). The revised failure of F3 thus turns on resampling, not on a difference between the two walks, whereas the registered half-width (0.049) is about five times the margin, and the registered F3 upper end reaches zero or above in 11 of the 50 seeds. The spender interval ends vary little across seeds (standard deviations of 0.0002 to 0.0015), and Rule B fails in every seed of both analyses because F2 does. The labels are sparse on the spender side: 1,495 of the 7,960 revised W1 spenders gained a new approval pair and 1,371 a new sender (2,850 and 2,812 of 34,587 in the registered analysis), so each spender coefficient rests largely on how those subsets are ordered.

VII. SYBIL EXPOSURE OF THE AUTHORIZATION LAYER

The results above concern honest data. A reputation layer must also be judged by what it costs to fake. This section calibrates the farm results of Gyöngyi and Garcia-Molina [35] to EndorseRank's authorization walk with uniform restarts, on its graph of the revised analysis at the freeze of 30 June 2026 (2,513,982 nodes and 3,808,949 edges), with the registered graph for comparison. The closed-farm bound (9) holds for any uniformly restarted walk on this layer, C-PR at $\lambda = 1$ included, but the star counts below are read against the cohort's EndorseRank scores. The model was computed after the W1 labels were known and enters no rule.

A. COST OF AN EDGE

Approving a spender reserves nothing: no tokens are locked, and the approving address may hold none of the token. Under bounded weights an approval of a few hundred dollars weighs almost as much as an unlimited one, and after row normalization a single out-edge carries the whole row whatever its amount. An edge between two attacker addresses therefore exposes nothing and costs only the fee. Because the revised graphs keep only the approvals set in the twelve months before each freeze, an approval leaves both EndorseRank and the authorization layer of C-PR a year after it was set, so a farm re-emits its approvals at least once a year; inside a closed farm the age discount of EndorseRank does not matter, because each row of a farm address is normalized over edges that all stay in the farm. Over q such periods a closed farm costs

$$C = q\left(|E_{\mathcal{S}}|\, c_{\text{gas}} + c_{\text{rev}}\right), \tag{8}$$

where c_{gas} is the fee of one approval and c_{rev} a per-period allowance for revocations or detection avoidance. We did not measure c_{gas}; whatever its value, the cost is linear in edges and periods. Restricting the graph to five widely traded tokens does not raise this cost, since approving one of them costs a farm no more than approving any other token; what it removes are logs that a token contract writes on behalf of owners who never acted.

B. WHAT A CLOSED FARM COLLECTS

Let every node of a farm $\mathcal{S}$ have out-edges, all inside $\mathcal{S}$, and let no honest owner approve a farm address. Write r_D for the stationary mass on dangling nodes, all of which are honest. Summing (3) over $\mathcal{S}$ with uniform $\boldsymbol{\pi}$ gives

$$\sum_{i\in\mathcal{S}} r_i = \frac{|\mathcal{S}|}{|V|}\cdot\frac{1-d+d\,r_D}{1-d} \le \frac{|\mathcal{S}|}{(1-d)|V|+d|\mathcal{S}|}, \tag{9}$$

where the bound uses $r_D \le 1-\sum_{i\in\mathcal{S}} r_i$. The second factor is the multiple of its population share that the farm captures, and it exceeds one whenever dangling nodes exist, because their redistributed mass reaches the farm. On the revised graph $r_D = 0.365$, 0.394 and 0.419 at $d = 0.75$, 0.85 and 0.95, so a closed farm captures 2.10, 3.23 and 8.96 times its population share; a ring of 20 addresses, each approving the next, inserted into the graph captures 3.23 times its share at $d = 0.85$. On the registered graph the factors are 2.01, 2.96 and 6.62.

C. CONCENTRATING THE MASS

The bound limits a farm's total mass, but the attacker can concentrate it. In a star, f farm addresses each approve a target t and t approves each of them back. With $c = (1-d+d\,r_D)/|V|$, the mass each node receives from jumps and dangling rows, the balance equations give

$$r_t = \frac{c\,(1+d\,f)}{1-d^2}. \tag{10}$$

On the revised graph $c = 1.93\times10^{-7}$, the common score of every node without in-edges. With $f = 10$, eleven addresses and twenty approvals per period, the target scores 6.60×10^{-6} on the graph with the star added, the value of (10), above the median of the cohort's scores (5.82×10^{-7}) and about two fifths of their 90th percentile (1.56×10^{-5}). Each further farm address adds $d\,c/(1-d^2)$ at the price of two approvals per period. Solving (10) for f shows what each rank costs. The median takes no star at all, because a pair of addresses that approve each other already reaches $c/(1-d)$. The 90th percentile takes 26 farm addresses, the 99th (1.35×10^{-3}) about 2,292, and the highest score among the cohort's contracts

(0.0278) about 47,073, a count that ignores how a farm that large would itself enlarge $|V|$ and lower c. On the registered graph the numbers are 13, 1,253 and about 1,080,271. A farm thus reaches the upper tenth of the cohort's scores with 52 approvals per period (26 on the registered graph), whose fee we did not measure, while the very top takes a farm more than a thousand times larger, because the leading contracts hold a large share of the stationary mass.

D. THE SAME ARGUMENT FOR THE COUPLED WALK

In C-PR at any $\lambda > 0$, the renormalization in (5) lets a node whose out-edges are all approvals follow them with weight one. A farm whose addresses approve only one another and make no payments is therefore closed in C-PR exactly as in the authorization walk, and (9) applies with C-PR's dangling mass. We did not compute that mass or run the farms on C-PR's graph, so the multipliers above are not claimed for it; the argument shows only that coupling does not close the gap. In the capped C-PR layer a purchased approval from an honest owner shifts much mass only when it is large relative to that owner's other allowances, and an approval at the cap lets the spender draw at least U dollars from the owner. In EndorseRank's bounded layer, by contrast, a fresh approval of a few hundred dollars takes 68.3% of the median owner's row, because most owners' latest approvals are old or small.

E. REMEDY, UNTESTED

The asymmetry that Cheng and Friedman identify points to the fix [36]. When jumps and the mass of dangling nodes are confined to a vetted seed set, as in TrustRank [14], a farm outside the seeds gains mass only along edges from honest owners, and those edges must be bought. We did not evaluate seeded restarts, and S-PR is not such a remedy, because its seeds are authorization scores that a farm can inflate. Credential gates such as Human Passport [26] address a different question, whether an address belongs to a distinct person, which has no direct counterpart for a contract, and they carry their own weaknesses [40]. The analysis is model-based and was checked with the solver on the observed graph; no attack was run on chain.

VIII. THREATS TO VALIDITY

A. CONSTRUCT VALIDITY

The spender labels are arrivals of the same kinds of edge that build the layers, so they measure future attachment and favor scores that track current counts; neither is a credit, safety or loss outcome. The trader label is the outcome closest to default in these data, but forced liquidation on one exchange is not default on a loan, and only 9 of the 70 W1 traders had a liquidation. The authorization edge is the latest approval up to the anchor, not the allowance in force (Section III); in the revised analysis an allowance set before the twelve-month score window does not count, however long it stays in force, and reading allowances from contract state at the freeze block would settle how many pairs counted as new were old relationships re-emitted. The layer weights matter: in the plan the raw-amount layers let unlimited approvals and high-decimal tokens dominate rows, which the dollar amounts and the cap of the revised analysis prevent. The two analyses reached the same verdicts on rules A and B, but they differ in weighting, tokens and score windows at once and are not a sweep over any of them, and other couplings, such as layers with EndorseRank's and AWP's bounded weights, are not evaluated in this paper.

B. INTERNAL VALIDITY

Rule A was fixed before any log of this study was extracted. Rule B's hypothesis came from an exploratory W0 contrast, and its margin from a W0 interval. The registration is a commit in the authors' own repository, not an entry in an independent registry, and its precedence over the W1 extraction rests on commit and job times rather than on a third-party timestamp (Section IV-C). The revised analysis, with its dollar amounts, five tokens and twelve-month score windows, was defined after every registered result was known, so its choices could favor one outcome. Two facts limit that risk without removing it: its slopes and cap come from quantiles of events in the twelve months up to the W0 freeze, so no event of a label window enters them, and it reached the same verdicts as the registered analysis on rules A and B. Reporting it first is likewise a choice made after the results; the registered analysis stays in every table and figure of results as the test that the rules were fixed for. The rule that scores absent contracts as zero is a pipeline choice; S2 shows that it changes F5 in the registered analysis through the tie structure but no part of the decision.

C. STATISTICAL CONCLUSION VALIDITY

Rule-bound intervals come from 400 paired resamples, as fixed in the plan; their Monte Carlo error is small against the distances that decided the verdicts, except for the revised F3, whose lower end sits at the margin and clears it in one of 50 further seeds (Appendix A). Rule B's verdict does not depend on that seed, because F2 fails in every one. The decision is an intersection of three conditions, so no multiplicity correction applies to it, but the many contrasts reported beside it are nominal and should not be read as a family of tests. The contract bootstrap treats spenders as independent although one new owner can approve several spenders. The trader cohort is small, and the margin taken from the spenders was at the edge of what the traders allow in the revised analysis and out of reach in the registered one (Section VI-I).

D. EXTERNAL VALIDITY

The evidence comes from one rollup, its contracts approved by at least three owners, scores built from the year before each freeze (from every event since October 2023 in the registered analysis) and two three-month label windows. Each graph is the one-hop ego network of the cohort, so the out-strengths and walk paths belong to the ego network, not to the whole chain; deeper samples, in which endorsements travel more than one step, could behave differently. Only

fixed global weights with uniform teleportation were tested; node-dependent or learned weights, mixtures of stationary vectors, interlayer copies of nodes and per-layer damping remain open.

IX. IMPLICATIONS FOR LENDING AND MARGIN PROTOCOLS

A protocol that screens counterparties or integrations from on-chain relations [17], [51] can take four practical points from these results. First, the cheapest score was hard to beat on its own construct: counting the distinct owners that approve a contract anticipated its later approvals better than every walk in this study, the coupled walk included, and in the revised analysis counting the distinct senders anticipated its later payers better than AWP and the coupled walk, with no solver and no restart vector to defend. The count is no harder to fake: approvals from N fresh addresses raise it by N, for N approval fees. Second, mixing an authorization layer into a payment walk moved the ranking mainly by switching the authorization layer on, and the gain on later approvals came with a loss on later payments larger than the registered margin; a protocol that wants both signals can report the two counts, or AWP and an approval count, side by side rather than fold them into one stationary distribution. Third, any score that walks authorization edges with uniform restarts can be inflated by a closed farm that needs no tokens, only approval fees, which we did not measure; on the revised graph about two dozen farm addresses would reach the top tenth of the cohort's EndorseRank scores (26, against 13 on the registered graph), and since the graph holds only the approvals of the past year, the farm keeps its edges by re-emitting them once a year. Restricting restarts to vetted seeds is the remedy the theory points to, but we did not test it. Finally, approval logs are not allowance state: a protocol that acts on allowances should read them from contract state at the decision block rather than reconstruct them from a window of events.

X. CONCLUSION

Authorization and payment give two readings of the same contracts. On this sample a walk that blends them at every node ranked later approvals better than either layer alone, though only slightly better than the authorization walk, but ranked later payers worse than the payment walk, so it did not beat both layers on both labels. In the main analysis, with US-dollar amounts on five tokens and a one-year score window before each freeze, defined after the results under the plan were known and applied as a re-analysis under the same rules, C-PR did no worse than the authorization walk on later approvals in the first window but did not beat the payment walk on later payers, so the rule fixed in the plan before this study extracted any log was not met. In the window registered before this study extracted it, C-PR repeated its gain over the payment walk on later approvals ($+0.170$ $[0.149, 0.189]$), but it ranked later payers worse by more than the registered margin (-0.025 $[-0.038, -0.015]$), against AWP by more still, and on the liquidation-free trading of contract traders it was level with the payment walk ($+0.000$ $[-0.010, 0.010]$) but too imprecise to stay within the margin. The pre-specified analysis in raw token units, the test that both rules were written for, reached the same verdicts ($+0.101$ $[0.092, 0.111]$, -0.025 $[-0.029, -0.021]$ and -0.044 $[-0.100, -0.002]$ on the three registered criteria). The gain on approvals appears as soon as the authorization layer is switched on and changes little across interior weights, and seeding the payment walk with authorization scores recovered only part of it. In contrasts made afterwards, C-PR ranked both spender labels above EndorseRank, and its lead over AWP on later approvals came largely from contracts that AWP scores zero. Several of those contrasts went against it: it ranked later approvals below the raw count of approving owners in both windows and below EndorseRank with activity restarts in the second, later payers below the raw count of senders and below AWP in both windows, and the traders, on their liquidation-free close rate, below the approval count. Under uniform restarts the authorization layer can also be inflated with a few dozen approvals a year, and coupling does not remove that exposure. Future work should fix in advance, anchor with a third-party timestamp and test on unextracted data couplings with comparable weights on both layers, samples deep enough for multi-hop propagation, trader cohorts large enough for the margin, and seed-restricted restarts under explicit attack.

APPENDIX A EXPLORATORY CHECKS

All checks below were run after every result of both analyses was known, by the script in the supplement, which reads the published per-contract score files and summaries and builds no graph. None of them enters a rule. Table 6 lists the results.

Recomputation. From the score files the script recomputes A1, A2, F1–F6 and S1 in both analyses, with the pipeline's resampling and SciPy's τ_b; every recomputed value equals the published one. S2 cannot be recomputed, because the isolated-node scores of the single-layer walks are not in the score files.

Seed spread. Across 50 further seeds of 400 resamples, the verdicts of A1, A2, F1 and F2 do not change in either analysis, and Rule B fails in every seed (Table 6(a)). On the spender contrasts the interval ends have standard deviations of 0.0002 to 0.0015; on the trader contrast F3 they move more. The revised F3 lower end ranges from -0.0146 to -0.0098 and clears the margin in one of the 50 seeds, so its narrow failure at the registered seed rests on resampling and not on a difference between the walks, whose coefficients are equal. In the registered analysis F3 fails in every seed because its lower end stays near -0.11, but its upper end reaches zero or above in 11 of the 50 seeds, so the reading that C-PR ranked the traders worse than the payment walk depends on the seed; the failure of the rule does not. One of the 20,000 trader resamples of each analysis was skipped because the label was constant on it.

Steps of the weight. Switching the authorization layer on (λ from 0 to 0.25) raises τ_b on new approval pairs by 0.092 to 0.169 and lowers it on new senders by 0.016 to 0.024,

TABLE 6. Exploratory Checks, Computed After Every Result of Both Analyses Was Known

		Revised analysis	Registered analysis
(a) Decision contrasts under 50 further bootstrap seeds of 400 resamples: lower interval end, mean (SD) [range]; seeds whose verdict differs from seed 42			
A1, W0, new approval pairs	upper end	0.0415 (0.0006) [0.0400, 0.0431]; 0 of 50	0.0132 (0.0004) [0.0124, 0.0142]; 0 of 50
A2, W0, new transfer senders	lower end	−0.0399 (0.0006) [−0.0411, −0.0388]; 0 of 50	−0.0336 (0.0002) [−0.0342, −0.0329]; 0 of 50
F1, W1, new approval pairs	lower end	0.1507 (0.0015) [0.1479, 0.1531]; 0 of 50	0.0926 (0.0006) [0.0905, 0.0939]; 0 of 50
F2, W1, new transfer senders	lower end	−0.0352 (0.0008) [−0.0376, −0.0339]; 0 of 50	−0.0281 (0.0003) [−0.0287, −0.0275]; 0 of 50
F3, W1, liquidation-free close rate	lower end	−0.0121 (0.0011) [−0.0146, −0.0098]; 1 of 50	−0.1064 (0.0044) [−0.1173, −0.0969]; 0 of 50
(b) Steps of the weight: τ_b of C-PR at the higher λ minus at the lower one, paired, 400 resamples, seed 42			
W0 spenders, new approval pairs	$0 \to 0.25$	+0.151 [0.132, 0.170]	+0.092 [0.083, 0.101]
	$0.25 \to 0.75$	−0.002 [−0.005, 0.001]	+0.002 [0.000, 0.003]
	$0.75 \to 1$	−0.029 [−0.040, −0.018]	−0.008 [−0.013, −0.003]
W0 spenders, new transfer senders	$0 \to 0.25$	−0.021 [−0.030, −0.012]	−0.024 [−0.027, −0.020]
	$0.25 \to 0.75$	−0.018 [−0.021, −0.015]	−0.013 [−0.015, −0.012]
	$0.75 \to 1$	−0.060 [−0.071, −0.048]	−0.095 [−0.102, −0.087]
W1 spenders, new approval pairs	$0 \to 0.25$	+0.169 [0.149, 0.187]	+0.099 [0.090, 0.109]
	$0.25 \to 0.75$	+0.001 [−0.002, 0.004]	+0.002 [0.001, 0.003]
	$0.75 \to 1$	−0.020 [−0.031, −0.009]	−0.010 [−0.014, −0.006]
W1 spenders, new transfer senders	$0 \to 0.25$	−0.016 [−0.029, −0.007]	−0.020 [−0.023, −0.016]
	$0.25 \to 0.75$	−0.017 [−0.020, −0.014]	−0.012 [−0.013, −0.010]
	$0.75 \to 1$	−0.058 [−0.069, −0.045]	−0.095 [−0.102, −0.088]
W1 traders, liquidation-free close rate	$0 \to 0.25$	+0.000 [−0.007, 0.008]	−0.037 [−0.085, −0.001]
	$0.25 \to 0.75$	+0.002 [−0.010, 0.014]	−0.013 [−0.036, 0.002]
	$0.75 \to 1$	+0.018 [−0.091, 0.125]	+0.222 [−0.062, 0.487]
(c) Rank agreement: τ_b between the C-PR scores at two values of λ over the cohort, 400 resamples, seed 42			
Matched cohort, end of window	$\lambda = 0$ and 0.25	0.587 [0.569, 0.602]	0.624 [0.616, 0.632]
	$\lambda = 0.25$ and 0.75	0.879 [0.875, 0.884]	0.869 [0.867, 0.871]
	$\lambda = 0.75$ and 1	0.596 [0.585, 0.610]	0.618 [0.612, 0.624]
W0 spenders	$\lambda = 0$ and 0.25	0.557 [0.542, 0.572]	0.598 [0.592, 0.605]
	$\lambda = 0.25$ and 0.75	0.889 [0.886, 0.893]	0.884 [0.882, 0.886]
	$\lambda = 0.75$ and 1	0.659 [0.648, 0.670]	0.658 [0.652, 0.663]
W1 spenders	$\lambda = 0$ and 0.25	0.558 [0.543, 0.573]	0.594 [0.586, 0.600]
	$\lambda = 0.25$ and 0.75	0.882 [0.879, 0.886]	0.884 [0.882, 0.886]
	$\lambda = 0.75$ and 1	0.622 [0.612, 0.634]	0.660 [0.655, 0.665]

Computed from the per-contract score files of the pipeline after every result of both analyses was known; no row enters a rule. The script first recomputes every decision contrast of Table 5 except those of S2 and reproduces the published values exactly. In (a) the A1 row reports the upper end, which decides ''no worse''; every other row the lower end.

with intervals that exclude zero in both windows and both analyses (Table 6(b)). Between $\lambda = 0.25$ and 0.75 the change is at most 0.002 in either direction on new approval pairs and between -0.012 and -0.018 on new senders. Dropping the payment layer (λ from 0.75 to 1) lowers both spender labels, by 0.008 to 0.029 on new approval pairs and by 0.058 to 0.095 on new senders. On the trader label only the registered step from 0 to 0.25 has an interval that excludes zero (-0.037 $[-0.085, -0.001]$); the registered step from 0.75 to 1 is $+0.222$ $[-0.062, 0.487]$, the revised one $+0.018$ $[-0.091, 0.125]$.

Rank agreement. The C-PR rankings at $\lambda = 0.25$ and $\lambda = 0.75$ agree at τ_b between 0.869 and 0.889 on every cohort of both analyses, with intervals no wider than 0.009, whereas each end point agrees with its nearest interior value at 0.557 to 0.624 ($\lambda = 0$ and 0.25) and 0.596 to 0.660 ($\lambda = 0.75$ and 1) (Table 6(c)). The interior is flat on the labels because the ranking itself changes little between interior weights, while the two end points are different operators.

ACKNOWLEDGMENT

The authors thank the School of Computing at the University of South Africa for supporting the first author's doctoral research, of which this study is part. The event logs were drawn from the public Arbitrum One dataset hosted on Google BigQuery; the study's 56 BigQuery jobs billed USD 42.24 in total. During the preparation of this work the authors used Claude (Anthropic) in order to assist with language editing, restructuring and checking of the text and analysis scripts. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

AUTHOR CONTRIBUTIONS

Taehong Kwon: conceptualization, methodology, software, data curation, formal analysis, writing – original draft. Ernest Mnkandla: supervision, writing – review & editing. Donatien Koulla Moulla: supervision, writing – review & editing. David Sena Attipoe: supervision, writing – review & editing.

CONFLICT OF INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

DATA AVAILABILITY

The study uses only public blockchain data, public daily token prices and the public Chainlink feed directory, and involves no human participants. The code, configuration, plan, registration, revision documents, extraction manifest, cost ledger, per-contract score files and machine-readable result summaries with their SHA-256 checksums are in the public repository https://github.com/abitocodes/phd_works, folder `1-Dissertation-Works/ERC20-Allowance-PageRank-Wallet-Reputation/experiments` (named `ERC20-Allowance-PageRank-Contract-Reputation/experiments` in the first seven commits of Table 3); the registered results are in `data/3-published-results-for-thesis/` and the revised ones in its subfolder `revised-usd-12m/`, committed in `950a7a4` and computed at code revision `0a94215`. The extracted graph tables are kept in the authors' BigQuery project and can be rebuilt from the public log table with the committed queries. This manuscript's source, its table and figure scripts and its supplement (the queries that extract and aggregate the data, and the exploratory-check script and its output) are in https://github.com/abitocodes/phd-dissertation-draft-en, folder `manuscripts/B-ieee-access`. The scores rank contracts only; externally owned accounts enter the graphs as owners, senders and recipients, no off-chain identity was collected or linked, and the paper reports aggregate results only.

REFERENCES

[1] F. Schär, "Decentralized finance: On blockchain- and smart contract-based financial markets," *Federal Reserve Bank of St. Louis Review*, vol. 103, no. 2, pp. 153–174, 2021, doi: 10.20955/r.103.153-74.

[2] L. Gudgeon, D. Perez, D. Harz, B. Livshits, and A. Gervais, "The decentralized financial crisis," in *2020 Crypto Valley Conference on Blockchain Technology (CVCBT)*. IEEE, 2020, pp. 1–15, doi: 10.1109/CVCBT50464.2020.00005.

[3] S. Werner, D. Perez, L. Gudgeon, A. Klages-Mundt, D. Harz, and W. Knottenbelt, "SoK: Decentralized finance (DeFi)," in *Proceedings of the 4th ACM Conference on Advances in Financial Technologies*. ACM, 2022, pp. 30–46, doi: 10.1145/3558535.3559780.

[4] H.-D. Do and T. Do, "PageRank and HodgeRank on Ethereum transactions: A measure for social credit," *International Journal of Software Innovation*, vol. 11, no. 1, pp. 1–13, 2023, doi: 10.4018/IJSI.315737.

[5] T. Do, D. Do, and L. Nguyen, "On-chain reputation ranking by Adaptive Weighted PageRank," in *2023 RIVF International Conference on Computing and Communication Technologies (RIVF)*. IEEE, 2023, pp. 318–323, doi: 10.1109/RIVF60135.2023.10471809.

[6] D. Lin, J. Wu, Q. Fu, Z. Zheng, and T. Chen, "RiskProp: Account risk rating on Ethereum via de-anonymous score and network propagation," *IEEE Transactions on Dependable and Secure Computing*, vol. 22, no. 3, pp. 2132–2145, 2025, doi: 10.1109/TDSC.2024.3475825.

[7] E. Bellini, Y. Iraqi, and E. Damiani, "Blockchain-based distributed trust and reputation management systems: A survey," *IEEE Access*, vol. 8, pp. 21 127–21 151, 2020, doi: 10.1109/ACCESS.2020.2969820.

[8] N. G. Packin and Y. Lev-Aretz, "Decentralized credit scoring: Black box 3.0," *American Business Law Journal*, vol. 61, no. 2, pp. 91–111, 2024, doi: 10.1111/ablj.12240.

[9] F. Vogelsteller and V. Buterin, "ERC-20: Token standard," Ethereum Improvement Proposals, Ethereum Improvement Proposal 20, 2015. [Online]. Available: https://eips.ethereum.org/EIPS/eip-20

[10] M. Lundfall, "ERC-2612: Permit extension for EIP-20 signed approvals," Ethereum Improvement Proposals, Ethereum Improvement Proposal 2612, 2020. [Online]. Available: https://eips.ethereum.org/EIPS/eip-2612

[11] D. Wang, H. Feng, S. Wu, Y. Zhou, L. Wu, and X. Yuan, "Penny wise and pound foolish: Quantifying the risk of unlimited approval of ERC20 tokens on Ethereum," in *Proceedings of the 25th International Symposium on Research in Attacks, Intrusions and Defenses*. ACM, 2022, pp. 99–114, doi: 10.1145/3545948.3545963.

[12] A. Halu, R. J. Mondragón, P. Panzarasa, and G. Bianconi, "Multiplex PageRank," *PLoS ONE*, vol. 8, no. 10, 2013, art. no. e78293, doi: 10.1371/journal.pone.0078293.

[13] M. De Domenico, A. Solé-Ribalta, E. Omodei, S. Gómez, and A. Arenas, "Ranking in interconnected multilayer networks reveals versatile nodes," *Nature Communications*, vol. 6, 2015, art. no. 6868, doi: 10.1038/ncomms7868.

[14] Z. Gyöngyi, H. Garcia-Molina, and J. Pedersen, "Combating web spam with TrustRank," in *Proceedings of the Thirtieth International Conference on Very Large Data Bases*. Morgan Kaufmann, 2004, pp. 576–587, doi: 10.1016/B978-012088469-8.50052-8.

[15] M. G. Kendall, "A new measure of rank correlation," *Biometrika*, vol. 30, no. 1/2, pp. 81–93, 1938, doi: 10.2307/2332226.

[16] GMX, "GMX docs," Accessed: Oct. 9, 2026. [Online]. Available: https://docs.gmx.io/

[17] D. Perez, S. M. Werner, J. Xu, and B. Livshits, "Liquidations: DeFi on a knife-edge," in *Financial cryptography and data security: FC 2021, Part II*, ser. Lecture Notes in Computer Science, vol. 12675. Springer, 2021, pp. 457–476, doi: 10.1007/978-3-662-64331-0_24.

[18] T. Kwon, E. Mnkandla, D. Koulla Moulla, and D. S. Attipoe, "Allowance edges or transfer edges? Evaluating on-chain smart contract reputation out of window against degree baselines," 2026, unpublished manuscript.

[19] L. Page, S. Brin, R. Motwani, and T. Winograd, "The PageRank citation ranking: Bringing order to the web," Stanford InfoLab, Technical Report SIDL-WP-1999-0120, 1999. [Online]. Available: http://ilpubs.stanford.edu:8090/422/

[20] A. N. Langville and C. D. Meyer, *Google's PageRank and beyond: The science of search engine rankings*. Princeton University Press, 2006, doi: 10.1515/9781400830329.

[21] J. M. Kleinberg, "Authoritative sources in a hyperlinked environment," *Journal of the ACM*, vol. 46, no. 5, pp. 604–632, 1999, doi: 10.1145/324133.324140.

[22] S. D. Kamvar, M. T. Schlosser, and H. Garcia-Molina, "The EigenTrust algorithm for reputation management in P2P networks," in *Proceedings of the 12th International Conference on World Wide Web*. ACM, 2003, pp. 640–651, doi: 10.1145/775152.775242.

[23] J. Wu, Q. Yuan, D. Lin, W. You, W. Chen, C. Chen, and Z. Zheng, "Who are the phishers? Phishing scam detection on Ethereum via network embedding," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 52, no. 2, pp. 1156–1166, 2022, doi: 10.1109/TSMC.2020.3016821.

[24] X. F. Liu, X.-J. Jiang, S.-H. Liu, and C. K. Tse, "Knowledge discovery in cryptocurrency transactions: A survey," *IEEE Access*, vol. 9, pp. 37 229–37 254, 2021, doi: 10.1109/ACCESS.2021.3062652.

[25] F. Victor, "Address clustering heuristics for Ethereum," in *Financial cryptography and data security: FC 2020*, ser. Lecture Notes in Computer Science, vol. 12059. Springer, 2020, pp. 617–633, doi: 10.1007/978-3-030-51280-4_33.

[26] Human Passport, "Human Passport developer platform," Accessed: Oct. 9, 2026. [Online]. Available: https://docs.passport.human.tech/

[27] T. H. Haveliwala, "Topic-sensitive PageRank: A context-sensitive ranking algorithm for web search," *IEEE Transactions on Knowledge and Data Engineering*, vol. 15, no. 4, pp. 784–796, 2003, doi: 10.1109/TKDE.2003.1208999.

[28] G. Jeh and J. Widom, "Scaling personalized web search," in *Proceedings of the 12th International Conference on World Wide Web*. ACM, 2003, pp. 271–279, doi: 10.1145/775152.775191.

[29] A.-L. Barabási and R. Albert, "Emergence of scaling in random networks," *Science*, vol. 286, no. 5439, pp. 509–512, 1999, doi: 10.1126/science.286.5439.509.

[30] M. E. J. Newman, "Clustering and preferential attachment in growing networks," *Physical Review E*, vol. 64, no. 2, 2001, art. no. 025102, doi: 10.1103/PhysRevE.64.025102.

[31] D. Liben-Nowell and J. Kleinberg, "The link-prediction problem for social networks," *Journal of the American Society for Information Science and Technology*, vol. 58, no. 7, pp. 1019–1031, 2007, doi: 10.1002/asi.20591.

[32] D. Lin, J. Wu, Q. Xuan, and C. K. Tse, "Ethereum transaction tracking: Inferring evolution of transaction networks via link prediction," *Physica A: Statistical Mechanics and its Applications*, vol. 600, 2022, art. no. 127504, doi: 10.1016/j.physa.2022.127504.

[33] S. Fortunato, M. Boguñá, A. Flammini, and F. Menczer, "Approximating PageRank from in-degree," in *Algorithms and models for the web-graph:*

WAW 2006, ser. Lecture Notes in Computer Science, vol. 4936. Springer, 2008, pp. 59–71, doi: 10.1007/978-3-540-78808-9_6.
[34] J. R. Douceur, "The Sybil attack," in *Peer-to-peer systems*, ser. Lecture Notes in Computer Science, vol. 2429. Springer, 2002, pp. 251–260, doi: 10.1007/3-540-45748-8_24.
[35] Z. Gyöngyi and H. Garcia-Molina, "Link spam alliances," in *Proceedings of the 31st International Conference on Very Large Data Bases*. VLDB Endowment, 2005, pp. 517–528. [Online]. Available: https://www.vldb.org/conf/2005/papers/p517-gyongyi.pdf
[36] A. Cheng and E. Friedman, "Sybilproof reputation mechanisms," in *Proceedings of the 2005 ACM SIGCOMM Workshop on Economics of Peer-to-Peer Systems*. ACM, 2005, pp. 128–132, doi: 10.1145/1080192.1080202.
[37] F. Victor and A. M. Weintraud, "Detecting and quantifying wash trading on decentralized cryptocurrency exchanges," in *Proceedings of the Web Conference 2021*. ACM, 2021, pp. 23–32, doi: 10.1145/3442381.3449824.
[38] L. W. Cong, X. Li, K. Tang, and Y. Yang, "Crypto wash trading," *Management Science*, vol. 69, no. 11, pp. 6427–6454, 2023, doi: 10.1287/mnsc.2021.02709.
[39] D. Siddarth, S. Ivliev, S. Siri, and P. Berman, "Who watches the watchmen? A review of subjective approaches for Sybil-resistance in proof of personhood protocols," *Frontiers in Blockchain*, vol. 3, 2020, art. no. 590171, doi: 10.3389/fbloc.2020.590171.
[40] A. A. Bordeianu and D. E. Popescu, "Threat landscape in decentralized systems: Sybil attacks, related vulnerabilities, and blockchain security evolution (2015–2025)," *Applied Sciences*, vol. 16, no. 14, 2026, art. no. 6929, doi: 10.3390/app16146929.
[41] B. A. Nosek, C. R. Ebersole, A. C. DeHaven, and D. T. Mellor, "The preregistration revolution," *Proceedings of the National Academy of Sciences*, vol. 115, no. 11, pp. 2600–2606, 2018, doi: 10.1073/pnas.1708274114.
[42] L. J. Cronbach and P. E. Meehl, "Construct validity in psychological tests," *Psychological Bulletin*, vol. 52, no. 4, pp. 281–302, 1955, doi: 10.1037/h0040957.
[43] J. Leskovec, A. Rajaraman, and J. D. Ullman, *Mining of massive datasets*, 3rd ed. Cambridge University Press, 2020, doi: 10.1017/9781108684163.
[44] DefiLlama, "DefiLlama API documentation: Coins endpoints," Accessed: Oct. 10, 2026. [Online]. Available: https://api-docs.defillama.com/
[45] Chainlink, "Selecting quality data feeds," Chainlink documentation, Accessed: Oct. 10, 2026. [Online]. Available: https://docs.chain.link/data-feeds/selecting-data-feeds
[46] H. Kalodner, S. Goldfeder, X. Chen, S. M. Weinberg, and E. W. Felten, "Arbitrum: Scalable, private smart contracts," in *Proceedings of the 27th USENIX Security Symposium*. USENIX Association, 2018, pp. 1353–1370. [Online]. Available: https://www.usenix.org/conference/usenixsecurity18/presentation/kalodner
[47] L. T. Thibault, T. Sarry, and A. S. Hafid, "Blockchain scaling using rollups: A comprehensive survey," *IEEE Access*, vol. 10, pp. 93 039–93 054, 2022, doi: 10.1109/ACCESS.2022.3200051.
[48] V. Buterin, S. Wilson, A. Dietrichs, and lightclient, "EIP-7702: Set code for EOAs," Ethereum Improvement Proposals, Ethereum Improvement Proposal 7702, 2024. [Online]. Available: https://eips.ethereum.org/EIPS/eip-7702
[49] B. Efron, "Bootstrap methods: Another look at the jackknife," *The Annals of Statistics*, vol. 7, no. 1, pp. 1–26, 1979, doi: 10.1214/aos/1176344552.
[50] T. J. DiCiccio and B. Efron, "Bootstrap confidence intervals," *Statistical Science*, vol. 11, no. 3, pp. 189–228, 1996, doi: 10.1214/ss/1032280214.
[51] M. Bartoletti, J. H.-y. Chiang, and A. Lluch Lafuente, "SoK: Lending pools in decentralized finance," in *Financial cryptography and data security: FC 2021 international workshops*, ser. Lecture Notes in Computer Science, vol. 12676. Springer, 2021, pp. 553–578, doi: 10.1007/978-3-662-63958-0_40.

TAEHONG KWON is a Ph.D. candidate in computer science with the School of Computing, University of South Africa, Johannesburg, South Africa. Research interests include on-chain reputation, decentralized finance and graph-based ranking.

ERNEST MNKANDLA is a Professor with the School of Computing, University of South Africa, Johannesburg, South Africa. Research interests include software engineering and artificial intelligence.

DONATIEN KOULLA MOULLA is with the Centre for Augmented Intelligence and Data Science (CAIDS), School of Computing, University of South Africa, Johannesburg, South Africa.

DAVID SENA ATTIPOE is with the Centre for Augmented Intelligence and Data Science (CAIDS), School of Computing, University of South Africa, Johannesburg, South Africa.